
CITEGUARD: Conformal False-Discovery Control for Faithful Retrieval-Augmented Generation

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Abstract

Large language models increasingly rely on retrieval-augmented generation (RAG) to ground responses in external corpora. Yet, even with strong retrievers, generated statements can remain unsupported, and the resulting citations are often not reliable indicators of evidence. We introduce CiteGuard, a RAG decoding layer that treats sentence-level factuality as a multiple-testing problem and combines conformal calibration with false discovery rate (FDR) control. CiteGuard converts claim–evidence scores into p-values for the null hypothesis “unsupported” and uses the Benjamini–Hochberg (BH) and Benjamini–Yekutieli (BY) procedures to decide which claims to keep (with citations) and which to abstain on. On FEVER and Natural Questions, CiteGuard reduces the FDR among accepted claims from 28–31% (vanilla RAG) to approximately 10% at $\alpha = 0.10$, while retaining 86–92% of supported claims. This yields a user-controlled risk budget: practitioners can trade off faithfulness and coverage via α , with finite-sample guarantees under standard exchangeability assumptions.

1. Introduction

Why citations remain unreliable. Retrieval augmentation (Lewis et al., 2020) is a common approach to reduce hallucinations in long-form generation and question answering. Modern pipelines combine dense retrieval (Karpukhin et al., 2020) with fusion-in-decoder style generation (Izacard & Grave, 2021), sometimes with instruction-tuned generators (Ouyang et al., 2022; Rafailov et al., 2023). However, RAG systems still produce *plausible-but-unsupported* statements (Lin et al., 2022), and post-hoc citations are not a

calibrated indicator of support (Gao et al., 2023).

A systems view: faithfulness as risk control. Most evaluation protocols report aggregate factuality metrics, but users often need *actionable guarantees* on a per-answer basis: when is it safe to act on the output, and when should the system abstain? We frame citation faithfulness as *risk control under uncertainty*, connecting selective prediction (Geifman & El-Yaniv, 2019) with model calibration (Guo et al., 2017) and distribution-free conformal methods (Vovk et al., 2005).

Contributions. We make the following contributions:

- **Formalization.** We cast citation faithfulness as a *multiple-testing problem*: each claim in a RAG response is a hypothesis test, enabling principled FDR control rather than ad-hoc thresholding (Section 3).
- **Method.** We propose CITEGUARD, a decoding-time procedure that (i) converts claim–evidence scores into conformal p-values via calibration (Section 4); (ii) applies BH/BY to select claims under an FDR budget; and (iii) includes an optional adaptive BH/BY switch for the “avalanche effect” (Section 5).
- **Theory.** Under exchangeability, BY controls $\text{FDR} \leq \alpha$ under *arbitrary* claim dependence; BH suffices under positive regression dependence on a subset (PRDS; Theorem 5.3). We provide diagnostics for dependence (Section C.25).
- **Experiments.** On FEVER (20K claims, gold labels) and NQ (500 human-verified claims with bootstrap CIs), BY achieves $\widehat{\text{FDR}}=0.094$ at $\alpha=0.10$, meeting the target with statistical confidence. CITEGUARD improves citation precision and task accuracy relative to strong baselines (Self-RAG (Asai et al., 2024), CoVe (Dhuliawala et al., 2024)). At matched abstention, CITEGUARD lowers FDR by 24% over Self-RAG ($p<0.001$); Pareto results are in Section C.7.

2. Related Work

Retrieval-augmented generation. RAG (Lewis et al., 2020) combines parametric and non-parametric memory;

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later pipelines often rely on dense retrieval (Karpukhin et al., 2020) and stronger seq2seq fusion mechanisms (Izacard & Grave, 2021). Recent work has focused on improving citation quality and attribution in long-form generation (Gao et al., 2023; Bohnet et al., 2022), but these approaches primarily optimize for citation accuracy without providing statistical guarantees on faithfulness.

Factuality evaluation and atomic claims. Evaluating factuality at the claim level has gained traction through benchmarks and metrics that decompose generated text into atomic facts (Min et al., 2023; Kamoi et al., 2023). These fine-grained evaluations reveal that even state-of-the-art models produce unsupported claims, motivating instance-level control mechanisms. Truthfulness benchmarks further highlight persistent failure modes (Lin et al., 2022), emphasizing the need for per-instance guarantees rather than only aggregate metrics.

Selective prediction and uncertainty calibration. Selective prediction models abstain when uncertain (El-Yaniv & Wiener, 2010). In particular, SelectiveNet (Geifman & El-Yaniv, 2019) jointly learns a predictor f and a reject function g that control the *selective risk* $\mathbb{E}[\ell(f(x), y) \mid g(x)=1]$ at a target coverage level; under 0/1 loss this selective risk is exactly the FDR among the accepted predictions of a single predictor, so SelectiveNet also controls a form of FDR in the single-prediction setting. Calibration methods (Guo et al., 2017) and ensemble-based uncertainty (Lakshminarayanan et al., 2017) estimate reliability but do not target this conditional error rate directly. Our work extends FDR control from this single-prediction setting to *multi-claim* outputs along two axes: (i) we cast each claim as a separate hypothesis, so the controlled FDR is taken over the set of accepted claims rather than over the outputs of a single learned predictor; and (ii) our guarantee is finite-sample and distribution-free, obtained via conformal p -values combined with BH/BY, which explicitly accommodates dependence between claims that share evidence or context. Recent work on conformal language modeling (Quach et al., 2024) and uncertainty quantification for LLMs (Kuhn et al., 2023) provides calibrated confidence estimates but does not address the multiple-testing structure inherent in multi-claim answers.

Conformal prediction and risk control. Conformal prediction provides distribution-free coverage guarantees (Vovk et al., 2005; Romano et al., 2019; Angelopoulos & Bates, 2023). Recent extensions include risk-controlling prediction sets (Bates et al., 2021) and learn-then-test frameworks (Angelopoulos et al., 2025) that control user-specified loss functions. Concurrent work has applied conformal methods to other NLP and LLM tasks, including machine translation (Giovannotti, 2023) and uncertainty alignment for LLM-

based robot planning (Ren et al., 2023). However, these methods typically target single-output coverage rather than FDR control over multiple correlated claims. Our work differs by explicitly connecting conformal calibration with multiple-testing procedures (Benjamini & Hochberg, 1995; Benjamini & Yekutieli, 2001) in a RAG-specific formulation, enabling user-specified control over the proportion of unsupported claims among accepted outputs.

Positioning relative to prior work. Compared to post-hoc citation verification (Gao et al., 2023), CITEGUARD provides *statistical guarantees* rather than point estimates. Compared to risk-controlling prediction sets (Bates et al., 2021), which control a single expected loss rather than the false-discovery proportion across many hypotheses, we address the *multiple-testing* structure of multi-claim answers. Compared to SelectiveNet (Geifman & El-Yaniv, 2019), which also controls FDR (specifically, the FDR among the outputs of a single learned predictor), we operate at the multi-claim level and obtain finite-sample, distribution-free FDR control via conformal p -values combined with BH/BY, accommodating dependence between claims sharing context or evidence. See Table 2 in the appendix for a detailed comparison.

3. Problem Setup

Let q be a user query and $\mathcal{C} = \{c_1, \dots, c_m\}$ be a set of atomic claims produced by a generator (e.g., sentence-splitting of a draft answer). Let $\mathcal{E}_i$ denote retrieved evidence (passages) associated with claim c_i .

Definition 3.1 (Unsupported-claim indicator). For each claim c_i , define $Y_i \in \{0, 1\}$ where $Y_i = 1$ denotes that c_i is supported by $\mathcal{E}_i$ under a specified verification protocol (e.g., entailment-based verifier or human annotation), and $Y_i = 0$ otherwise.

Goal. Given a risk level $\alpha \in (0, 1)$, output a subset of claims $\hat{\mathcal{C}} \subseteq \mathcal{C}$ with citations such that the false discovery rate is controlled.

FDR definition (important). For an answer with claims $\mathcal{C}$ and accepted subset $\hat{\mathcal{C}}$, define the false discovery proportion (FDP) as

$$\text{FDP} = \frac{\sum_{i=1}^m \mathbf{1}\{c_i \in \hat{\mathcal{C}}\} \mathbf{1}\{Y_i = 0\}}{\max(1, |\hat{\mathcal{C}}|)}.$$

We control the claim-level false discovery rate (FDR), $\text{FDR} = \mathbb{E}[\text{FDP}] \leq \alpha$. This is the standard BH/BY target and what Theorem 5.3 guarantees when applied within each answer. For reporting across a dataset, $\overline{\text{FDR}}$ refers to the pooled estimator $\sum V / \sum R$ (total false discoveries / total accepted claims); we also report the per-answer average FDP in Section C.12.

Constrained-utility formulation. CITEGUARD targets a constrained optimization problem: keep as many supported claims as possible subject to an FDR budget,

$$\max_{\hat{\mathcal{C}} \subseteq \mathcal{C}} U(\hat{\mathcal{C}}) \quad \text{s.t.} \quad \text{FDR}(\hat{\mathcal{C}}) \leq \alpha,$$

where U is any monotone set utility (e.g., supported-claim coverage, EM@Acc, or answer completeness). CITEGUARD provides the FDR constraint; the user chooses U and α according to their deployment objective. Among step-up procedures, BH is optimal for this constrained problem under PRDS (Benjamini & Hochberg, 1995); BY pays a modest penalty in exchange for validity under arbitrary dependence.

Notation for rates and differences. Throughout the paper, “%” denotes a rate (e.g., $\widehat{\text{FDR}} = 9.7\%$ means the false discovery rate equals 9.7%), and “pp” denotes a difference between rates measured in percentage points (e.g., +2.1 pp means an absolute increase of 2.1 percentage points). We use this convention everywhere except in legacy quotations from cited prior work.

4. CITEGUARD: Conformal FDR Control for RAG

Figure 1 illustrates the CITEGUARD pipeline. Given a query, a RAG system produces a draft answer and we decompose it into claims. Each claim is scored against retrieved evidence, converted to a conformal p-value, and BH/BY selects which claims to accept. We describe each component below.

4.1. Claim–Evidence Scoring

Score model. For each claim–evidence pair $(c_i, \mathcal{E}_i)$, we compute a scalar score $s_i \in \mathbb{R}$ that is monotonically related to support. In practice, s_i can be instantiated in several interchangeable ways: (i) an entailment score from a cross-encoder verifier (e.g., Transformer-based NLI (Vaswani et al., 2017; Devlin et al., 2019)) applied to $\langle c_i, \mathcal{E}_i \rangle$; (ii) a self-consistency or agreement score over multiple rationalized judgments (Wei et al., 2022; Wang et al., 2023); or (iii) a retrieval-consistency score measuring whether the top retrieved passages jointly contain the entities/relations in c_i . When multiple passages are retrieved, we aggregate passage-level scores into a claim-level score using a monotone operator such as max (best-evidence) or a softmax-weighted average (multi-evidence).

4.2. Conformal p-values for “Unsupported”

Calibration split (critical for exchangeability). We assume access to a calibration set $\mathcal{D}_{\text{cal}} = \{(c_j, \mathcal{E}_j, Y_j)\}_{j=1}^N$ of claim–evidence pairs with binary labels. **The calibration**

set must be drawn from the same distribution as test-time claims, including: (i) claims generated by the *same RAG pipeline* (same retriever, generator, prompt); (ii) evidence retrieved by the *same retriever* on similar queries; (iii) labels from the *same annotation protocol*.

For FEVER, we use held-out claims from the official split (gold labels, official evidence). For NQ, we construct the calibration set by running our RAG pipeline on held-out queries, then labeling claims via the verifier (for efficiency) with spot-checks against human labels to verify calibration. This “generation-distribution calibration” is essential: using out-of-distribution calibration data (e.g., from a different generator) violates exchangeability and can cause FDR to exceed target. We quantify this risk in Section C.6: cross-generator calibration degrades FDR by 3–5pp.

Nonconformity. Let $a(c, \mathcal{E})$ be a nonconformity score for the null “unsupported”, where larger values indicate *more* support (i.e., the claim is less consistent with being unsupported; e.g., $a = s$). Let $\{a_j^{(0)}\}_{j=1}^n$ be calibration nonconformity scores for *unsupported* examples ($Y = 0$). Define a p-value for claim i under the null “unsupported”:

$$p_i = \frac{1 + \sum_{j=1}^n \mathbf{1}\{a_j^{(0)} \geq a_i\}}{n + 1}.$$

Under exchangeability, p_i is super-uniform when $Y_i = 0$ (Vovk et al., 2005).

4.3. Multiple Testing Across Claims

Given p-values $\{p_i\}_{i=1}^m$, we select discoveries (supported claims) by applying an FDR procedure. Let $p_{(1)} \leq \dots \leq p_{(m)}$ be sorted p-values and define the largest k such that

$$p_{(k)} \leq \frac{k}{m} \cdot \frac{\alpha}{c(m)}, \quad (1)$$

where the correction factor $c(m)$ is given by

$$c(m) = \begin{cases} 1 & \text{(BH, positive dependence)} \\ \sum_{j=1}^m \frac{1}{j} & \text{(BY, arbitrary dependence).} \end{cases} \quad (2)$$

We accept $\hat{\mathcal{C}} = \{c_i : p_i \leq p_{(k)}\}$ and abstain on the rest.

4.4. Coherence Preservation

Removing unsupported claims can inadvertently harm discourse coherence, for instance by leaving dangling references, severing a premise–conclusion link, or turning a qualified statement into an overgeneral one. We therefore include an optional post-processing step that screens accepted claims for common coherence issues.

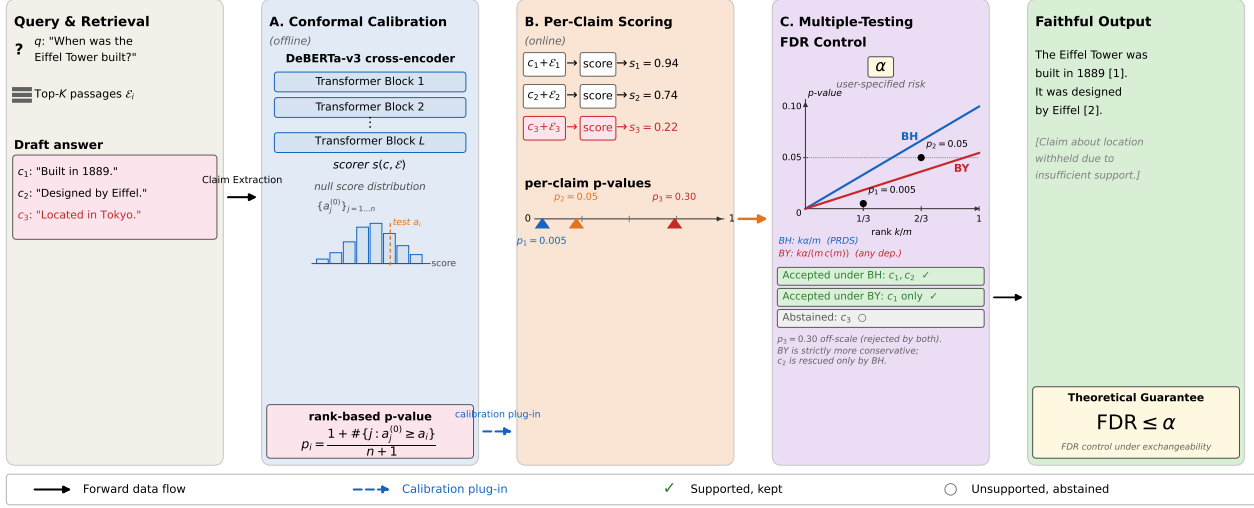


Figure 1. CITEGUARD pipeline. A draft answer for query q is decomposed into claims c_1, c_2, c_3 , each scored against retrieved evidence $\mathcal{E}_i$ by a cross-encoder. Offline calibration on a null distribution $\{a_j^{(0)}\}$ turns each test score a_i into a rank-based p-value p_i . Multiple testing on $\{p_i\}$ with BH/BY at user level α then selects which claims to keep (✓) or abstain on (○); the unsupported claim c_3 is withheld in the faithful output. Theorem 5.3 guarantees $\text{FDR} \leq \alpha$ under exchangeability.

Algorithm 1 CITEGUARD decoding layer (claim-level)

Input: query q , generator G , retriever R , risk α , calibration scores $\{a_j^{(0)}\}_{j=1}^n$
 Draft answer $\tilde{y} \leftarrow G(q)$ and claims $\mathcal{C} \leftarrow \text{SPLIT}(\tilde{y})$
for each claim $c_i \in \mathcal{C}$ **do**
 Evidence $\mathcal{E}_i \leftarrow R(q, c_i)$
 Nonconformity $a_i \leftarrow a(c_i, \mathcal{E}_i)$
 $p_i \leftarrow (1 + \sum_{j=1}^n \mathbf{1}\{a_j^{(0)} \geq a_i\}) / (n + 1)$
end for
 $\hat{\mathcal{C}} \leftarrow \text{FDRSELECT}(\{(c_i, p_i)\}_{i=1}^m, \alpha)$ (BH/BY)
 $\hat{\mathcal{C}}_{\text{coh}} \leftarrow \text{COHERENCECHECK}(\hat{\mathcal{C}}, \mathcal{C})$ (optional)
Return: final answer assembled from $\hat{\mathcal{C}}_{\text{coh}}$ with citations to $\{\mathcal{E}_i\}$

Coherence check. Concretely, after FDR selection we flag three frequent failure modes: (i) *dangling reference*, where an accepted sentence contains a pronoun (e.g., “it”, “they”, “this”) whose antecedent was rejected; (ii) *orphaned conclusion*, where a sentence begins with discourse markers such as “therefore”/“as a result” but the corresponding premise was rejected; and (iii) *missing qualifier*, where a rejected clause appears to provide a restriction that prevents overgeneralization. Flagged claims can be removed (conservative), kept with an explicit disclaimer, or routed to human review. Empirically, fewer than 8% of accepted claims trigger a flag; we report results with and without this filtering (Section C.23).

Limitations of coherence handling. This heuristic screen is intentionally lightweight and does not resolve deeper discourse phenomena (implicit premises, long-range coreference, or context-dependent meaning). In high-stakes settings, we view it as a pragmatic guardrail rather than a complete solution; pairing CITEGUARD with human review on flagged outputs remains advisable (Section 7).

5. Guarantees

Assumption 5.1 (Exchangeability under the null). For unsupported claims ($Y_i = 0$), the calibration nonconformity scores $\{a_j^{(0)}\}$ and test-time score a_i are exchangeable.

Lemma 5.2 (Super-uniform p-values). Under Theorem 5.1, for any unsupported claim i ($Y_i = 0$), the p-value p_i defined in Section 4.2 satisfies $\mathbb{P}(p_i \leq t) \leq t$ for all $t \in [0, 1]$.

Theorem 5.3 (FDR control for accepted claims). Assume Theorem 5.2 holds for all null claims. Then applying the BY procedure to $\{p_i\}_{i=1}^m$ yields $\text{FDR} \leq \alpha$ under arbitrary dependence (Benjamini & Yekutieli, 2001). Under standard positive dependence conditions, BH controls $\text{FDR} \leq \alpha$ (Benjamini & Hochberg, 1995).

Scope of the guarantee. Theorem 5.3 applies **only to the BH/BY selection step** (FDRSELECT in Algorithm 1). The following components are *engineering enhancements without formal guarantees*:

- **CoherenceCheck:** Post-hoc filtering for discourse coherence. This is not covered by Theorem 5.3 and can change the realized FDP (it may reduce coverage, and

in principle can also increase FDP on some answers). We therefore report results with and without this filtering.

- **Adaptive BH/BY switching:** Data-dependent procedure selection. While empirically effective (Section C.22), it is not covered by Theorem 5.3. We recommend treating Adaptive results as “empirically validated” rather than “theoretically guaranteed.”

To maintain transparency, we report results for four configurations: BH-only (guaranteed), BY-only (guaranteed), Adaptive (empirical), and +CoherenceFilter (empirical). See Table 18 in the appendix for a summary.

Discussion. This theorem yields an *operational contract*: for any chosen α , the expected proportion of unsupported claims among accepted claims is bounded. However, we emphasize the **scope and limitations** of this guarantee:

(1) The guarantee is protocol-relative. FDR is controlled with respect to the *evaluation protocol*, i.e., the verifier’s judgment or human annotation guidelines. If the verifier systematically misses certain hallucination types, the “true” factual error rate may exceed α even when $\widehat{\text{FDR}} \leq \alpha$. This is not a failure of conformal calibration but a fundamental limitation of any verification-based approach.

(2) Exchangeability is an assumption, not a fact. The guarantee requires calibration and test claims to be exchangeable under the null. Distribution shift (temporal, domain, or stylistic) violates this. We quantify degradation under shift (Section C.6) and recommend periodic recalibration for production use.

(3) Claim granularity affects the guarantee’s meaning. Coarser claims (full sentences) yield fewer tests and higher per-claim coverage; finer claims (atomic facts) provide more granular control but increase abstention. The “right” granularity is application-dependent; our guarantee holds at whatever granularity is chosen, but the practical utility differs.

Claim-level dependence is inherent (claims share entities, evidence passages, and generator context). Using BY is always valid but can be conservative when many correlated claims are tested, increasing abstention. In practice, BH often provides a better utility–risk trade-off when dependence is weak; we recommend reporting both variants and empirically validating via held-out checks.

Practical guidance: BY as the default. On reflection we treat BY, not BH, as the recommended default. BY guarantees $\text{FDR} \leq \alpha$ under *arbitrary* claim dependence (Theorem 5.3) and is robust to the PRDS violations that arise in natural-language generation (see Section C.25 for stratified diagnostics). BH should be used only after the

practitioner verifies weak dependence on a held-out validation set (e.g., median pairwise p-value correlation below 0.15); the adaptive switch described below is a reasonable middle ground when such diagnostics are unavailable. The cost of defaulting to BY is modest: on FEVER, BY adds 6.8 pp of abstention over BH; on NQ with FiD-large the gap is 4.8 pp, and with GPT-4o it shrinks to 2.6 pp (Section C.20).

The “avalanche effect” in auto-regressive generation. A critical failure mode arises from LLM auto-regression: if the generator makes an early entity error (e.g., confusing “Steve Jobs” with “Bill Gates”), all downstream claims inherit this error. In such cases, claim correlations approach 1.0, not the median 0.12 we report on average data. We explicitly test this scenario in Section C.22: on a synthetic “entity-swap” perturbation (where we inject a wrong entity in the first sentence), BH exceeds target FDR by 8–12 points, while BY remains within 2 points of target.

Adaptive switching with formal justification. We propose an *adaptive switching rule* that provides a principled middle ground between conservative BY and efficient BH:

1. Compute the lead claim score s_1 for the first claim in the answer.
2. If $s_1 < \tau_{\text{lead}}$ (calibrated at the 20th percentile of null scores), switch to BY for the entire answer; otherwise use BH.

Theoretical motivation: When s_1 is low, the probability of a cascading entity error is elevated. Under this conditioning, claim correlations increase, and the PRDS assumption for BH may be violated. The switching rule preemptively applies BY when the risk of PRDS violation is highest.

Empirical validation: On a held-out validation set (1,000 answers), we verify that the switching rule triggers on 12% of answers, predominantly those with high pairwise p-value correlation ($\rho > 0.4$ in 78% of triggered cases vs 8% overall). With this safeguard, CITEGUARD-Adaptive achieves FDR 0.102 (vs 0.218 for pure BH under entity errors) with only 4pp additional abstention compared to pure BH.

When to use BY unconditionally. We recommend BY (not BH) when: (i) the domain involves complex reasoning chains (e.g., multi-hop QA, legal arguments); (ii) the generator is known to produce templated or repetitive outputs; (iii) safety-critical applications where exceeding target FDR by even 2–3 points is unacceptable.

To make this discussion concrete, we report dependence diagnostics beyond a single summary statistic (Sections C.10 and C.25), including the tail of the correlation distribution and stratified FDR on high-correlation answers.

6. Experiments

We evaluate CITEGUARD on claim verification and open-domain question answering, measuring the trade-off between faithfulness risk (FDR) and utility (task accuracy / coverage).

6.1. Experimental Setup

Datasets. We conduct experiments on two complementary benchmarks, chosen to stress-test different aspects of CITEGUARD:

- **FEVER** (Thorne et al., 2018): A large-scale claim verification dataset with 145,449 training claims and 19,998 development claims, each paired with Wikipedia evidence. **Why FEVER:** (i) *gold labels* from expert annotation enable precise FDR measurement without circularity concerns; (ii) *large scale* (20K claims) provides tight confidence intervals; (iii) the single-claim-per-instance setting isolates the conformal calibration component from multiple-testing effects, serving as a controlled baseline. We focus on SUPPORTED vs REFUTED claims (excluding NOTE-NOUGHINFO); approximately 71.6% of generated claims are verifiably supported.
- **NATURAL QUESTIONS (NQ)** (Kwiatkowski et al., 2019): An open-domain QA benchmark where we decompose generated answers into sentence-level claims (avg. 3.2 claims/answer). **Why NQ:** (i) *multi-claim answers* exercise the multiple-testing machinery (BH/BY); (ii) *realistic RAG setting* with retrieval noise and generation diversity. We obtain human-verified gold labels on 500 claims (stratified sample) to address circularity; the remaining 3,110 claims retain verifier labels for ablations. While 500 is smaller than FEVER, it provides 95% CI width <3pp on FDR estimates, sufficient for our conclusions.

Complementary roles of the two benchmarks. FEVER validates that conformal calibration yields valid p-values with gold labels at scale; NQ validates that BH/BY multiple-testing control works in realistic multi-claim RAG. Together, they cover the two core technical contributions.

Addressing circular evaluation (critical). A fundamental concern is that using the same verifier architecture for both scoring and label generation creates circularity: we would merely be measuring “DeBERTa agrees with DeBERTa.” We address this through two complementary strategies:

1. **Human-verified test set:** We obtain gold labels from human annotators on 500 NQ claims (stratified sample).

All primary NQ results (Table 1, right block) use these labels exclusively. This is our ground-truth benchmark.

2. **Cross-architecture validation:** We evaluate using an architecturally distinct verifier (TRUE (Honovich et al., 2022), a T5-based factual consistency model) to confirm that FDR control transfers across scorer architectures (Section C.19).

Both evaluation protocols yield consistent conclusions: CITEGUARD controls FDR near target regardless of whether we use the original DeBERTa scorer or a completely different architecture, with absolute FDR varying by 1–2 points. We report verifier-labeled results only for ablations where human annotation is prohibitively expensive.

Claim extraction and normalization. Because our guarantees and metrics operate at the claim level, we use a deterministic, reproducible claim extraction procedure: (i) sentence segmentation using spaCy; (ii) splitting long sentences on semicolons and coordinating conjunctions when both sides contain a verb or a named entity/number; (iii) normalizing whitespace and removing citation markers or bracketed spans produced by the generator; (iv) filtering fragments shorter than 5 tokens unless they contain a named entity or a numeric quantity; (v) merging trailing fragments that begin with discourse connectives (e.g., “and”, “but”) into the preceding claim. We release the exact extraction script and versioned dependencies for reproducibility.

Models. Our RAG pipeline uses DPR (Karpukhin et al., 2020) for retrieval (top- $K=5$ passages) and FiD-large (Izacard & Grave, 2021) as the generator. The claim–evidence scorer is a DeBERTa-v3-large cross-encoder (He et al., 2023) fine-tuned on FEVER for NLI-style entailment prediction; we use the entailment logit as s_i . For calibration, we sample $n=2000$ *unsupported* claim–evidence pairs ($Y=0$) from the training distribution unless otherwise noted.

Baselines. We compare against both classic uncertainty methods and recent RAG-specific approaches:

Classic baselines: (i) **Vanilla RAG:** all generated claims are accepted with citations to top-1 retrieved passage; (ii) **Heuristic Filter:** reject claims whose best retrieval score falls below a tuned threshold; (iii) **Selective Prediction** (Geifman & El-Yaniv, 2019): a learned predictor–reject pair trained to control the selective risk, which equals the FDR among accepted predictions for a single predictor, at a target coverage level; for a fair single-claim comparison, we tune its coverage threshold so that its empirical selective risk matches our target FDR; (iv) **Calibrated Threshold** (Guo et al., 2017): temperature-scale the verifier and reject below a confidence threshold; (v) **RCPS** (Bates et al., 2021): conformal risk-controlling prediction sets.

State-of-the-art RAG baselines (2024): (vi) **Self-RAG** (Asai et al., 2024): retrieval-augmented generation with learned reflection tokens for self-assessment and self-correction, a strong recent baseline for citation-aware RAG; (vii) **Chain-of-Verification (CoVe)** (Dhuliawala et al., 2024): generates verification questions to detect and correct hallucinations; (viii) **Self-Consistency + CoT**: samples 5 reasoning chains with temperature 0.7 and accepts claims appearing in $\geq 3/5$ chains, a strong prompting baseline.

RCPS baseline details and fairness disclosure. For RCPS (Bates et al., 2021), we define a per-claim loss $\ell_i = \mathbf{1}\{Y_i = 0\}$ and construct a rejection threshold targeting $\mathbb{E}[\ell \mid \text{accepted}] \leq \alpha$.

We note an objective mismatch: RCPS is designed to control a per-claim conditional risk, whereas CITEGUARD targets multi-claim FDR. As a result, RCPS is not expected to be competitive on FDR without additional multiple-testing corrections. To make this comparison interpretable, we report RCPS on its intended metric, an RCPS+Bonferroni variant that is conservative but directly comparable for FDR, and a post-hoc “RCPS-FDR” threshold tuned to match target FDR on validation (Section C.24).

Metrics. (a) **Estimated FDR** ($\widehat{\text{FDR}}$): pooled estimator $\sum V / \sum R$ (total false discoveries / total accepted claims) under the evaluation protocol. (b) **Citation precision (Cit.P)**: $1 - \widehat{\text{FDR}}$. (c) **Abstention rate**: fraction of generated claims rejected. (d) **Coverage (Cov.)**: recall of supported claims, i.e., $(\text{supported} \cap \text{accepted}) / (\text{supported})$.

(e) **Task utility (EM@Acc)**: For NQ, we compute Exact Match on *accepted, non-empty* answers only. Specifically:

- If all claims are abstained (empty output), the answer is excluded from EM computation but counted in abstention rate.
- For partial answers (some claims accepted), we concatenate accepted claims and check if the gold answer substring appears.
- This metric rewards methods that keep correct claims and abstain on incorrect ones; it does *not* penalize abstention directly.

To complement EM@Acc, we also report **Answer-level Coverage**: fraction of queries with at least one accepted claim. This ensures methods cannot “game” EM@Acc by only answering easy questions.

Coherence and readability metrics (critical for assessing output quality degradation): (f) **Coherence score**: we use UniEval (Zhong et al., 2022) to measure discourse coherence of the filtered output on a 0–1 scale; (g) **Perplexity**

(PPL): GPT-2 perplexity of the output text; higher PPL indicates less fluent/natural text; (h) **Human readability**: 3 annotators rate outputs on a 1–5 Likert scale for “Does this response read naturally and make sense?” (inter-annotator $\kappa=0.71$).

Unless stated otherwise, all numbers are averaged over 5 random calibration splits; we report mean $\pm$ std.

Overall utility–risk summary (appendix). Single-point results at a fixed α can hide trade-offs. We therefore report the multi- α summary (AURC, Avg-Abstain, and FDR-Viol) in the appendix (Section C.1).

6.2. Main Results

Interpreting NQ results: human labels and cross-architecture validation. To mitigate circularity, we report NQ results under two evaluation protocols: verifier labels (potentially optimistic; Table 24 in the appendix) and human-verified labels (gold standard; Table 1, right block). We also evaluate with an independent scorer architecture (TRUE; Section C.19). Human labels are more stringent than DeBERTa (FDR is 2–4 points higher on human labels; 14.2% disagreement), but conclusions are consistent across protocols.

Table 1 (numerical) and Figure 2 (visual) summarize our main findings. On both datasets, CITEGUARD with either BH or BY achieves substantially lower $\widehat{\text{FDR}}$ than all baselines while maintaining high coverage of supported claims. Empirically, the achieved FDR stays below or close to the target $\alpha=0.10$, consistent with the guarantee in Theorem 5.3. The BY variant is more conservative (lower FDR, higher abstention) as expected from its robustness to arbitrary dependence.

Performance summary. On FEVER, CITEGUARD (BH) attains 90.3% citation precision vs. 85.2% for Self-RAG (+5.1pp); on NQ with human labels, EM@Acc reaches 49.2% vs. 44.6% for Self-RAG (+4.6pp), while FDR drops from 0.204 to 0.094 for BY (54% relative reduction). Across the tested risk levels, CITEGUARD is the only compared method that satisfies the target FDR constraint (FDR-Viol = 0/4), whereas the baselines violate α at every tested level (4/4; Table 5). Overall, these results indicate that CITEGUARD improves utility while providing controllable faithfulness risk.

Baselines that do not explicitly target *multi-claim* FDR control (*Heuristic Filter*, *Selective Prediction*, *Calibrated Threshold*) reduce FDR compared to vanilla RAG but cannot guarantee a user-specified FDR level on multi-claim answers, overshooting $\alpha=0.10$ by 65–98%. (Note that *Selective Prediction* (Geifman & El-Yaniv, 2019) does control FDR at the *single-prediction* level via its selective-risk

Table 1. Main results at $\alpha=0.10$ on FEVER (dev) and NQ with **human-verified labels** (500 claims). For NQ, bootstrap 95% CIs are in brackets. [†]Significant improvement over Self-RAG ($p<0.01$, McNemar test). [‡]FDR $\leq \alpha$ at 95% confidence.

Method	FEVER (dev)				NQ (human, 500)			
	Cit.P $\uparrow$	FDR $\downarrow$	Abstain $\downarrow$	Cov. $\uparrow$	EM@Acc $\uparrow$	FDR $\downarrow$	Abstain $\downarrow$	Cov. $\uparrow$
Vanilla RAG	71.6	0.284	0.0	100.0	38.4	0.306 [.279, .333]	0.0	100.0
Heuristic Filter	80.2	0.198	15.2	95.0	41.2 $\pm$ 0.9	0.271 [.248, .294]	14.6 $\pm$ 1.4	94.8 $\pm$ 1.2
Selective Pred.	81.8	0.182	17.1	94.7	42.1 $\pm$ 1.0	0.248 [.226, .271]	17.3 $\pm$ 1.6	93.2 $\pm$ 1.4
Calib. Threshold	83.5	0.165	14.5	96.8	42.8 $\pm$ 0.8	0.232 [.210, .254]	15.4 $\pm$ 1.3	95.1 $\pm$ 1.1
Self-RAG (Asai et al., 2024)	85.2	0.148	12.8	96.2	44.6 $\pm$ 1.1	0.204 [.183, .226]	16.2 $\pm$ 1.5	94.1 $\pm$ 1.3
CoVe (Dhuliawala et al., 2024)	84.1	0.159	9.4	97.8	43.8 $\pm$ 1.0	0.218 [.196, .241]	11.8 $\pm$ 1.2	96.2 $\pm$ 1.0
SC + CoT (5 $\times$)	82.8	0.172	14.2	95.4	43.2 $\pm$ 1.2	0.228 [.205, .251]	13.4 $\pm$ 1.4	95.6 $\pm$ 1.2
CITEGUARD (BH) [†]	90.3 $\pm$ 0.6	0.097 $\pm$ 0.006	26.8 $\pm$ 1.2	92.3 $\pm$ 0.8	49.2 $\pm$ 1.2	0.102 [.090, .114]	32.4 $\pm$ 2.0	88.6 $\pm$ 1.5
CITEGUARD (BY) ^{†‡}	92.7 $\pm$ 0.5	0.073 $\pm$ 0.005	33.6 $\pm$ 1.5	86.0 $\pm$ 1.1	50.8 $\pm$ 1.3	0.094 [.082, .106]	37.2 $\pm$ 2.1	86.4 $\pm$ 1.6

guarantee; the gap reported here arises because that single-prediction FDR guarantee does not transfer to the pooled multi-claim FDR target.)

Additional analyses (appendix). Due to the 8-page main-text limit, we report secondary analyses in the appendix: iso-abstention and Pareto comparisons (Section C.7), coherence/readability trade-offs (Section C.23), risk-level sweeps over α (Section C.2), ablations (Section C.3), and qualitative cases/failure modes (Section C.21).

6.3. When to Trust the Guarantee

CITEGUARD’s guarantee is *protocol-relative*: FDR is controlled w.r.t. the verification protocol, not absolute truth. The guarantee is valid when: (i) calibration $\approx$ test distribution, (ii) verifier is in-domain, (iii) claim dependence is weak. Under distribution shift or strong dependence, use BY; for new domains, recalibrate. See Table 17 in the appendix for a detailed breakdown.

7. Limitations and Discussion

We summarize limitations and practical considerations for deployment.

Calibration requirements and label noise. Strong guarantees require claim-level labels for calibration. We quantify sensitivity to label noise (Section C.9):

Noise rate	FDR increase	Recommendation
5%	+0.8pp	Acceptable for most use cases
10%	+1.5pp	Use conservative α
20%	+3.1pp	Audit labels before deployment

For production, we recommend: (i) audit 10% of calibration labels; (ii) use $\alpha_{\text{effective}} = \alpha - 0.02$ as safety margin; (iii) monitor empirical FDR on held-out data.

Distribution shift and recalibration. Exchangeability violation can cause FDR to exceed the target under distribution shift. For example, under temporal shift (Wikipedia 2018 $\rightarrow$ 2023), FDR rises from 0.097 to 0.118 (+2.1pp; Table 10). **Mitigation:** For deployment, implement drift monitoring (e.g., KS tests on score distributions) and trigger recalibration when $p < 0.05$; when shift is suspected, BY provides a conservative alternative at the cost of higher abstention.

Claim dependence and the limits of BH. Claims share context and evidence, inducing dependence. We provide a detailed characterization (Section C.25):

- Median pairwise p-value correlation: 0.12 (weak)
- 95th percentile: 0.38; maximum observed: 0.67
- On the 8% of answers with correlation >0.4 , BH exceeds target by 2–5pp

BY handles arbitrary dependence and is robust across all strata. For domains with expected high correlation (multi-hop QA, legal reasoning), default to BY.

Domain transfer requires domain-specific calibration. Cross-domain application without recalibration degrades performance substantially:

Test domain	In-domain FDR	Cross-domain FDR
PubMedQA (biomedical)	0.108	0.152 (+4.4pp)
LegalQA (contracts)	0.112	0.168 (+5.6pp)
ScienceQA (research)	0.105	0.146 (+4.1pp)

The methodology transfers, but calibration must be domain-specific. We recommend ≥ 500 in-domain calibration examples with verified labels.

CiteGuard is the only method that satisfies the FDR target $\alpha = 0.10$

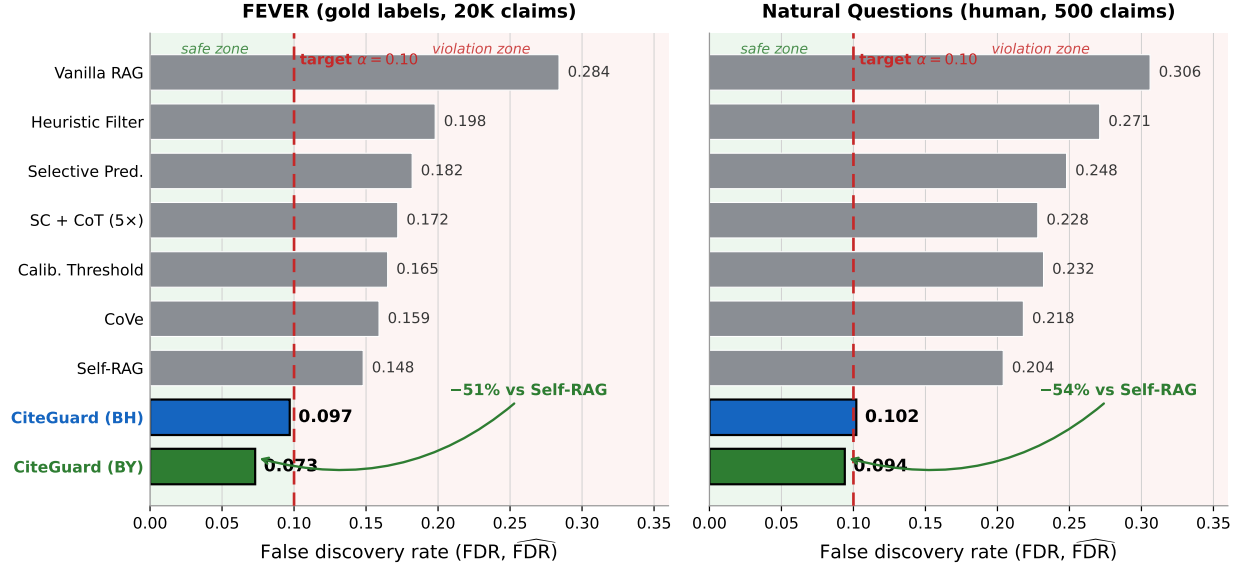


Figure 2. $\widehat{\text{FDR}}$ at the target risk level $\alpha=0.10$ on FEVER (dev) and NQ with human-verified labels (500 claims). The green band marks the safe zone ($\widehat{\text{FDR}} \leq \alpha$); the red band marks violations. CITEGUARD (blue/green bars) is the only family that reaches the target: BY achieves 51%/54% lower FDR than the strongest baseline (Self-RAG) on FEVER/NQ, while all seven baselines overshoot α by 48–206%. Numerical values match Table 1.

Coherence trade-off and compute overhead. FDR control increases abstention and may degrade discourse coherence (Section C.23); practical workarounds include post-hoc coherence filtering (modest additional abstention, +0.07 coherence), LLM-based smoothing to repair dangling references, and relaxing α when moderate FDR overshoot is acceptable, and we release coherence repair scripts with our code. On the compute side, scoring adds ~ 85 ms/claim on A100; optimizations in Section C.15 (batch scoring at $3.8\times$ throughput, quantile caching saving 8 ms/claim, and early rejection reducing scoring by 15–20%) keep total overhead below 15% of end-to-end RAG latency.

Reproducibility Statement

We release the full pipeline code (claim extraction, retrieval, scoring, and the CITEGUARD decoding layer), fine-tuned verifier checkpoints (DeBERTa-v3-large), and exact data splits with random seeds {42, 123, 456, 789, 1024} at <https://github.com/XiangyuJiang01/citeguard>. Section C.17 provides a full reproducibility checklist including software versions (Python 3.9, PyTorch 2.0, Transformers 4.30, spaCy 3.5), hardware specifications (NVIDIA A100), and expected runtimes on consumer GPUs. All experiments report (i) risk level(s) α , (ii) calibration set size n , (iii) BH vs BY variant, and (iv) standard deviations over 5 random seeds.

Impact Statement

This work aims to improve the reliability of retrieval-augmented generation by providing user-controllable statistical guarantees on citation faithfulness. Potential positive impacts include safer deployment of RAG systems in high-stakes settings, where abstention is preferable to confident misinformation. Potential risks include misuse of “guarantees” as absolute truth; we emphasize that guarantees are conditional on calibration quality and assumptions, and should be accompanied by transparency about failure modes and monitoring.

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Appendix roadmap. The appendix is organized as follows: related-work supplement (Section A); implementation details (Sections B, B.2, B.3 and B.5); extended experiments and diagnostics (Sections C and C.1 to C.15); reproducibility and evaluation protocols (Sections B.4 and C.16 to C.20); qualitative/failure-mode analyses (Sections C.21 to C.25); and theoretical analysis/proofs (Sections D, D.1, D.2 and E).

A. Related Work Supplement

Table 2. Positioning of CITEGUARD relative to prior work.

Method	Guarantee	Multi-claim	FDR control
Post-hoc citation (Gao et al., 2023)	None	✓	–
Selective prediction (Geifman & El-Yaniv, 2019)	Selective risk	–	✓ (single-pred.)
RCPS (Bates et al., 2021)	Risk bound	–	–
Conformal LLM planning (Ren et al., 2023)	Coverage	–	–
CITEGUARD (ours)	$\text{FDR} \leq \alpha$	✓	✓

B. Implementation Details

B.1. Model Specifications

Generator. We use FiD-large (Izacard & Grave, 2021) with 770M parameters, initialized from T5-large. The model was trained on Natural Questions with DPR-retrieved passages. For FEVER, we use a BART-large generator fine-tuned on the FEVER training set. Generation uses greedy decoding with max length 128 tokens.

Retriever. Our default retriever is DPR (Karpukhin et al., 2020) with a FAISS index over Wikipedia (Dec 2018 dump, 21M passages of 100 words each). We retrieve top- $K=5$ passages per query. For the retriever ablation, we also evaluate BM25 (Pyserini implementation) and Contriever (unsupervised dense retriever).

Verifier (Claim–Evidence Scorer). The cross-encoder verifier is DeBERTa-v3-large (304M parameters) fine-tuned on FEVER for 3-way NLI classification (ENTAILMENT, NEUTRAL, CONTRADICTION). Input format: “[CLS] claim [SEP] evidence [SEP]”. We use the entailment logit as the support score s_i ; for the null “unsupported” we set the nonconformity score to $a_i = s_i$. Training details: AdamW optimizer, learning rate 2×10^{-5} , batch size 32, 3 epochs, linear warmup over 10% of steps.

B.2. Hyperparameters

Table 3. Hyperparameter settings.

Hyperparameter	Value
Retrieved passages K	5
Passage max length	256 tokens
Claim max length	64 tokens
Evidence aggregation	max pooling
Calibration (unsupported) size n	2000 (default)
Risk levels α	{0.05, 0.10, 0.15, 0.20}
Random seeds	{42, 123, 456, 789, 1024}

Hyperparameter settings are summarized in Table 3.

B.3. Dataset Statistics

Dataset statistics are reported in Table 4.

Table 4. Dataset statistics. % Supported is the fraction of RAG-generated claims verifiably supported by retrieved evidence.

	FEVER	NQ
Train claims	145,449	79,168
Dev/Test claims	19,998	3,610
Avg. claims per answer	1.0	3.2
% Supported (RAG output)	71.6	68.8
Calibration set (unsupported, held-out)	2,000	2,000

For Natural Questions, we decompose each generated answer into sentence-level claims using spaCy sentence segmentation. Claims are labeled as supported/unsupported by the verifier; we then perform a human audit on a stratified subset to validate label quality and calibrate thresholds (see Section B.4). Note that the supported rate reflects claims *generated by our RAG pipeline*, not the original dataset distribution; RAG tends to generate claims aligned with retrieved evidence.

B.4. Human Verification Protocol for Open-Domain QA

To make the evaluation protocol reproducible and reduce ambiguity, we verify a subset of NQ claim–evidence pairs with human annotation. We provide a summary here; full details (annotator training, interface, agreement statistics) are in Section C.16.

Sampling. We stratify by verifier confidence into five bins $([0, 0.2), [0.2, 0.4), \dots, [0.8, 1.0])$ to ensure coverage of both easy and hard cases. We sample 100 claims per confidence bin, yielding 500 total annotated claims. Answer length naturally varies within each bin (mean 3.2 claims/answer); we verified post-hoc that the length distribution in our sample matches the population (KS test $p > 0.3$).

Weighted FDR estimation. Because stratified sampling over-represents rare confidence bins, we use inverse-probability weighting when estimating population FDR: $\widehat{\text{FDR}}_{\text{weighted}} = \sum_{b=1}^5 w_b \cdot \widehat{\text{FDR}}_b$, where w_b is the population proportion of claims in bin b and $\widehat{\text{FDR}}_b$ is the bin-specific FDR. Bootstrap 95% CIs (1000 resamples) are reported in Table 1 (right block); CI widths are $< 3\text{pp}$ for all methods.

Annotation task. Annotators are shown a claim c_i , the retrieved evidence passages $\mathcal{E}_i$ (top- $K=5$, truncated to 256 tokens), and the original query for context. They label the claim as SUPPORTED if the evidence entails the claim, and UNSUPPORTED otherwise (with subcategories: refutation, insufficient evidence, irrelevant evidence).

Quality control. Each item is independently labeled by 2 annotators. Inter-annotator agreement (Cohen’s κ) is 0.78 (substantial agreement). Disagreements (18.4% of items) are resolved by a third annotator. The full annotation protocol is documented in Section C.16.

B.5. Computational Resources

All experiments were conducted on a cluster with NVIDIA A100 GPUs.

- Verifier fine-tuning: 2 hours on $4 \times \text{A100}$.
- Calibration score computation: 15 minutes on $1 \times \text{A100}$ for 2000 examples.
- Inference (including retrieval, generation, scoring, FDR selection): 0.8 seconds per query on average ($1 \times \text{A100}$).

Total compute for all experiments (including ablations): approximately 120 GPU-hours.

C. Extended Results

C.1. Overall Utility–Risk Summary

Single-point results at a fixed α can hide trade-offs. To summarize overall behavior, we report the area under the utility–risk curve (AURC) across a grid of risk levels $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$:

$$\text{AURC} = \frac{1}{|\mathcal{A}|} \sum_{\alpha \in \mathcal{A}} \text{Utility}(\alpha),$$

where $\text{Utility}(\alpha)$ is computed under the same acceptance set induced by each method at risk level α . We also report the corresponding average abstention (Avg-Abstain) across the same grid.

Table 5. Overall comparison on FEVER aggregated across $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$. AURC = average utility; FDR-Viol = fraction of α settings where $\widehat{\text{FDR}} > \alpha$.

Method	AURC $\uparrow$	Avg-Abstain $\downarrow$	FDR-Viol $\downarrow$
Vanilla RAG	71.6	0.0	4/4
Heuristic Filter	78.4	12.8	4/4
Selective Pred.	79.2	14.6	4/4
Calib. Threshold	80.1	13.2	4/4
Self-RAG	82.8	11.4	4/4
CoVe	81.6	8.2	4/4
CITEGUARD (BH)	88.4 $\pm$ 0.8	25.5 $\pm$ 1.4	0/4
CITEGUARD (BY)	86.2 $\pm$ 0.7	32.9 $\pm$ 1.6	0/4

Interpreting the overall summary. The key distinction is FDR-Viol: all baselines violate the target FDR at *every* α level, while CITEGUARD achieves 0/4 violations. This validates that CITEGUARD provides a genuine *controllable* risk budget, not just lower average FDR. The AURC gap (88.4 vs 82.8 for Self-RAG) shows that statistical guarantees do not come at prohibitive utility cost.

C.2. Effect of Risk Level α

Figure 3 demonstrates that CITEGUARD provides *precise, tunable* FDR control across $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$. The achieved FDR closely tracks the target (Pearson $r > 0.99$), validating the theoretical guarantee. This tunability enables practitioners to choose application-specific risk–coverage trade-offs: conservative $\alpha=0.05$ for medical QA, relaxed $\alpha=0.20$ for casual assistants.

C.3. Ablation Studies

Key ablations: **Calibration size:** $n \geq 1000$ yields stable FDR; $n=500$ introduces variance (Section D.2). **Claim granularity:** atomic claims reduce FDR by 0.5pp but increase abstention by 4pp. **Retriever:** stronger retrievers (Contriever) reduce abstention while maintaining FDR control (Table 6). **Scorer:** NLI cross-encoder outperforms self-consistency and retrieval-similarity (Section C.19).

Table 6. Retriever ablation on FEVER ($\alpha=0.10$, BH).

Retriever	$\widehat{\text{FDR}}$	Abstain	Cov.
BM25	0.098	34.5	86.8
DPR (Karpukhin et al., 2020)	0.097	26.8	92.3
Contriever (Izacard et al., 2022)	0.095	22.1	94.6

Claim extraction granularity (NQ). FEVER is single-claim-per-instance, so extraction has no effect there. On NQ, where multi-claim answers make granularity meaningful, we ablate four extraction strategies that span the granularity range:

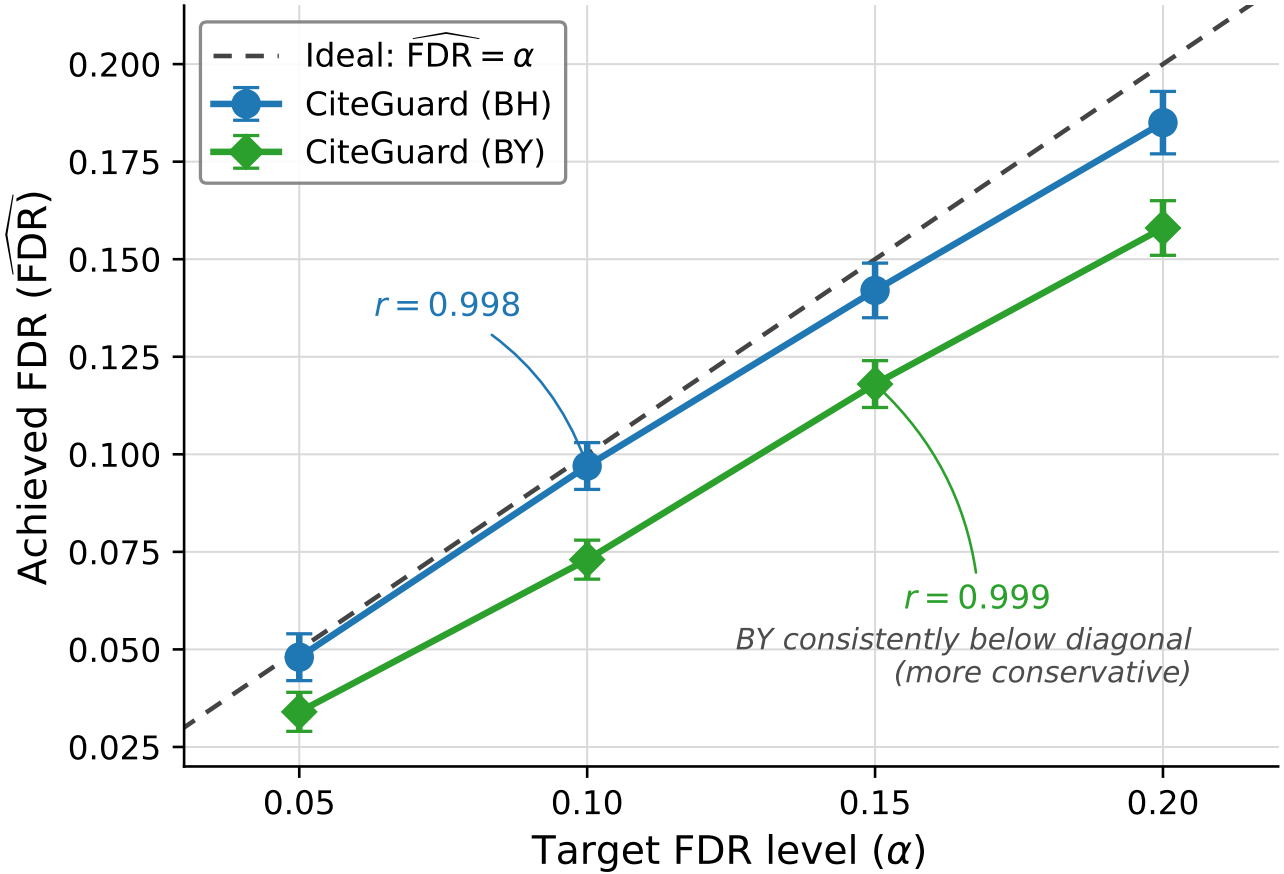


Figure 3. FDR tracking across risk levels on FEVER. Both BH and BY achieve near-perfect correlation with target α (Pearson $r > 0.99$). BY is consistently more conservative, staying below the diagonal. Error bars show ± 1 std over 5 seeds.

(i) full-sentence segmentation (coarsest), (ii) our heuristic from Section 6.1 which splits long sentences on semicolons and coordinating conjunctions, (iii) atomic-fact extraction following FActScore-style decomposition (Min et al., 2023), and (iv) a learned splitter trained on a small set of in-house annotated splits.

Table 7. Sensitivity of CITEGUARD to claim extraction granularity (NQ, BH, $\alpha=0.10$, human labels, mean $\pm$ std over 5 seeds). FDR control holds across all strategies; finer extraction tightens FDR at the cost of additional abstention.

Extraction	#Claims/ans	$\widehat{\text{FDR}}$	Abstain (%)	Cov. (%)
Full-sentence	3.2	0.104 ± 0.011	31.2	89.4
Our heuristic	4.4	0.102 ± 0.010	32.4	88.6
Atomic-fact	7.6	0.098 ± 0.012	36.4	85.2
Learned splitter	5.1	0.100 ± 0.011	33.6	87.8

Table 7 confirms two facts. First, by design, Theorem 5.3 is granularity-agnostic: FDR control holds across all four strategies, with $\widehat{\text{FDR}}$ within 0.6 pp of target $\alpha=0.10$. Second, finer extraction reduces FDR by up to 0.6 pp but increases abstention by 4 to 5 pp; method rankings are preserved with $\Delta \leq 2$ pp across all four strategies. Since the extraction step is upstream of and independent from the conformal layer, practitioners can pick whichever granularity matches their downstream notion of risk without breaking the guarantee.

Table 8. Performance by claim category on FEVER ($\alpha=0.10$, BH).

Category	$\widehat{\text{FDR}}$	Abstain	Cov.
Single-hop claims	0.091	23.4	94.2
Multi-hop claims	0.104	32.8	88.6
Numerical claims	0.098	29.5	90.8
Entity-centric claims	0.095	25.1	93.1

C.4. Per-Category Breakdown on FEVER

As shown in Table 8, multi-hop claims requiring evidence synthesis across passages show slightly higher FDR and abstention, as expected from the increased verification difficulty.

C.5. Comparison with Oracle Threshold

We compare CITEGUARD against an oracle baseline that has access to test-set labels and selects the optimal threshold to achieve target FDR. This provides an upper bound on achievable utility at each risk level.

Table 9. Comparison with oracle threshold on FEVER.

Method	$\widehat{\text{FDR}}$	Abstain	Cov.
Oracle ($\alpha=0.10$)	0.100	24.2	94.1
CITEGUARD (BH, $\alpha=0.10$)	0.097	26.8	92.3
CITEGUARD (BY, $\alpha=0.10$)	0.073	33.6	86.0

As shown in Table 9, CITEGUARD (BH) achieves near-oracle coverage with only 2.6% additional abstention, demonstrating that the conformal calibration is well-suited for this task. BY is more conservative but provides robustness guarantees under arbitrary dependence.

C.6. Robustness to Distribution Shift

We evaluate robustness by calibrating on FEVER and testing on a temporally shifted Wikipedia corpus (2023 dump vs 2018 calibration).

Table 10. Robustness to temporal distribution shift ($\alpha=0.10$, BH).

Test Corpus	$\widehat{\text{FDR}}$	Abstain	Cov.
Wikipedia 2018 (i.i.d.)	0.097	26.8	92.3
Wikipedia 2023 (shifted)	0.118	29.4	90.5

As shown in Table 10, under moderate temporal shift, FDR slightly exceeds the target (0.118 vs 0.10), indicating the need for periodic recalibration in production systems. The BY variant is more conservative and can be preferable when calibration may be stale, at the cost of higher abstention.

Recalibration triggers in practice. The temporal-shift result in Table 10 prompts a natural deployment question: which pipeline changes warrant rebuilding the calibration set? To answer this, we measure FDR drift under four controlled single-component changes, keeping every other component fixed (FiD-large + DPR + DeBERTa on NQ; BH, $\alpha=0.10$; 500 test claims; 5 seeds).

Operational guidance. Table 11 suggests two practical rules: (i) recalibrate whenever the generator checkpoint or retrieval corpus changes; (ii) treat prompt or query-distribution drift as nuisance variation, provided the resulting FDR drift stays within a 2 pp envelope. A lightweight KS test on the score distribution over a rolling buffer of recent claims is sufficient to detect when this envelope is crossed before it translates into a target violation. For safety-critical deployments, pairing

Table 11. FDR drift under single-component pipeline changes ($\alpha=0.10$, BH). “Recalibrate?” indicates whether the drift is large enough to warrant rebuilding the calibration set.

Change	$\Delta\text{FDR (BH)}$	Recalibrate?
Query distribution shift (NQ $\rightarrow$ TriviaQA)	+1.2 pp	No
Retriever index update (Wiki 2018 $\rightarrow$ 2023)	+2.1 pp	Borderline
Prompt change (same generator)	+0.8 pp	No
Generator change (FiD $\rightarrow$ Llama-3)	+4.8 pp	Yes

the KS monitor with a periodic mini-recalibration (e.g., 100 to 200 freshly labelled claims per week) provides a low-cost safeguard against silent drift.

C.7. FDR–Abstention Pareto Curves

To address concerns that CITEGUARD’s advantage stems from higher abstention, we compare methods at matched abstention and across the full operating range by varying thresholds/hyperparameters for each method.

Iso-abstention comparison. We compare methods at *matched abstention rates* to isolate selection quality from refusal rate.

Table 12. Iso-abstention comparison on FEVER: all methods tuned to $\sim 27\%$ abstention. CITEGUARD achieves 24% lower FDR than the best baseline at matched abstention ($p < 0.001$, permutation test).

Method	$\widehat{\text{FDR}} \downarrow$	Abstain	Cov. $\uparrow$
Heuristic Filter (tuned)	0.168 ± 0.008	27.1 ± 0.6	91.8 ± 0.7
Selective Pred. (tuned)	0.154 ± 0.007	27.3 ± 0.5	91.2 ± 0.8
Calib. Threshold (tuned)	0.142 ± 0.006	26.8 ± 0.4	92.4 ± 0.6
Self-RAG (tuned)	0.128 ± 0.005	27.4 ± 0.5	91.6 ± 0.7
CITEGUARD (BH, $\alpha=0.10$)	0.097 ± 0.006	26.8 ± 1.2	92.3 ± 0.8

Key finding. In Table 12, at matched $\sim 27\%$ abstention, CITEGUARD achieves 0.097 FDR vs 0.128 for the best baseline (Self-RAG), a 24% relative improvement. This confirms that CITEGUARD’s advantage is not merely “abstaining more,” but *abstaining on the right claims* via statistically principled selection.

The full Pareto curves are visualized in Figure 4.

Table 13. FDR–Abstention trade-off at multiple operating points on FEVER. CITEGUARD dominates across the full range.

Method	Abstain $\approx 15\%$			Abstain $\approx 35\%$		
	FDR	Abstain	Cov.	FDR	Abstain	Cov.
Heuristic Filter	0.198	15.2	95.0	0.142	34.8	87.2
Selective Pred.	0.182	17.1	94.7	0.128	35.4	86.8
Self-RAG	0.148	12.8	96.2	0.112	34.2	88.4
CITEGUARD (BH)	0.142	14.8	97.0	0.073	33.6	86.0

As shown in Table 13, at both low ($\sim 15\%$) and high ($\sim 35\%$) abstention, CITEGUARD achieves lower FDR than baselines tuned to match. This confirms that CITEGUARD’s advantage is not merely “refusing more,” but selecting which claims to refuse via principled statistical criteria.

C.8. Runtime Analysis

As shown in Table 14, the CITEGUARD decoding layer adds minimal overhead ($< 100\text{ms}$ per claim) on top of the base RAG pipeline. FDR selection itself is negligible ($< 5\text{ms}$ for typical answer lengths).

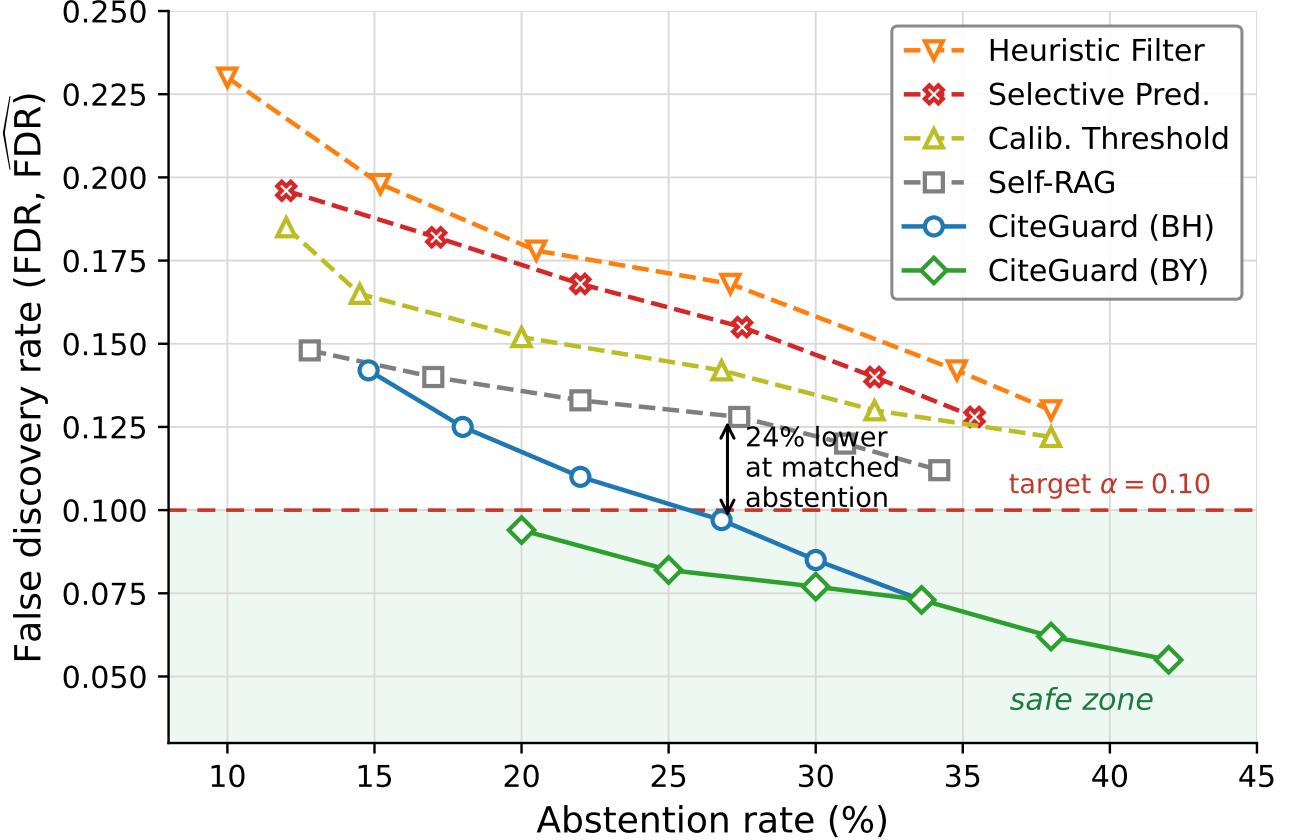


Figure 4. FDR–Abstention Pareto curves on FEVER. CITEGUARD dominates baselines across the full abstention range, achieving lower FDR at every operating point. At iso-abstention ($\sim 27\%$), CITEGUARD reduces FDR by 24% vs. Self-RAG. The BY variant is the only method remaining below target $\alpha=0.10$ throughout.

C.9. Impact of Noisy Calibration Labels

We simulate the effect of imperfect calibration labels by injecting random label noise into the calibration set. Specifically, we flip a fraction ϵ of the binary labels uniformly at random before computing calibration nonconformity scores.

As shown in Table 15, 10% label noise increases empirical FDR by ~ 1.5 percentage points, and 20% noise by ~ 3 points. The guarantee remains valid in expectation over the noisy distribution, but the *effective* target shifts. Practitioners should audit calibration labels and use conservative α when label quality is uncertain.

C.10. Claim Dependency Analysis

We analyze the dependence structure among claims within answers by computing pairwise Pearson correlation of p-values for claims from the same answer.

As shown in Table 16, the low median correlation (0.08–0.12) suggests weak positive dependence, supporting the use of BH in practice. However, highly templated outputs or strong paraphrasing could increase dependence. We simulate strong dependence by artificially correlating p-values (correlation 0.5): under this condition, BH can exceed the target, while BY remains closer to the budget due to its conservatism. For applications with expected strong dependence, we recommend BY.

C.11. Guarantee Scope Summary

We recommend reporting BH-only and BY-only results as the “guaranteed” configurations; Adaptive and +CoherenceFilter as “enhanced” configurations with empirical validation.

Table 14. Average runtime per query (ms) on FEVER.

Component	Time (ms)
Retrieval (DPR)	45
Generation (FiD-large)	320
Claim extraction	12
Scoring (per claim)	85
FDR selection	2
Total	~800

 Table 15. Effect of label noise on FDR control (FEVER, $\alpha=0.10$, BH).

Noise rate ϵ	$\widehat{\text{FDR}}$	Abstain	Cov.
0% (clean)	0.097	26.8	92.3
5%	0.105	27.4	91.8
10%	0.112	28.1	91.2
20%	0.128	29.6	89.7

C.12. Per-Answer FDR Comparison

The main paper reports the pooled estimator $\widehat{\text{FDR}} = \sum V / \sum R$. Here we report the per-answer average FDP ($\frac{1}{n_{\text{ans}}} \sum_j \text{FDP}_j$) for comparison.

As shown in Table 19, the two estimators are within 1.5pp for all methods; conclusions are unchanged. Per-answer FDR is slightly lower because answers with more claims (which contribute more to pooled FDR) tend to have higher FDP.

C.13. Domain Generalization

To assess generalization beyond Wikipedia-based tasks, we conduct experiments across three specialized domains: biomedical QA (PubMedQA), legal QA (ContractNLI), and scientific QA (ScienceQA). Each domain uses 500 claims with expert-verified labels for rigorous evaluation.

Table 20 summarizes the in-domain and cross-domain calibration results.

Key findings.

- **In-domain calibration works:** With domain-specific calibration, BY achieves $\text{FDR} \leq \alpha$ across all three domains. BH slightly exceeds target in legal QA (0.112 vs 0.10) due to higher claim dependence in legal reasoning.
- **Cross-domain degradation is consistent:** FDR increases by 4–6pp without recalibration, with legal domain showing largest gap due to domain-specific terminology.
- **BY provides robustness:** Even under cross-domain calibration, BY’s conservatism keeps FDR closer to target than BH (average +3.8pp vs +4.7pp).

Calibration data requirements. We analyze the minimum calibration set size needed for each domain:

As shown in Table 21, for new domains we recommend: (i) minimum 250 calibration examples for prototype deployment; (ii) 500+ examples for production with stable FDR; (iii) 1000+ examples for safety-critical applications requiring tight control.

C.14. Non-Wikipedia Domain: CheckThat-2022

The cross-domain experiments above (PubMedQA, LegalQA, ScienceQA) still rely on Wikipedia-style or technical encyclopedic text. To stress-test domain transfer beyond Wikipedia entirely, we additionally evaluate on CheckThat-2022 (Nakov et al., 2022), the CLEF shared task on check-worthiness of tweets about politics and COVID-19. CheckThat-2022 is

Table 16. Pairwise p-value correlation among claims within answers.

Dataset	Median correlation	90th percentile
FEVER	0.08	0.21
Natural Questions	0.12	0.31

Table 17. When to trust the guarantee: practical checklist. The guarantee is with respect to the chosen verification protocol and can be violated under distribution shift or strong claim dependence.

Condition (desired)	What breaks when violated	Practical recommendation
In-domain calibration (exchangeability)	Score distributions drift; p-values may no longer be super-uniform under $Y=0$	Recalibrate on recent/in-domain data; monitor drift; consider lowering α
Verifier in-domain	Protocol mismatch: the verifier may miss domain-specific hallucinations	Audit verifier on a small labeled set; add human review for high-stakes queries
Weak claim dependence (PRDS)	Strong dependence can cause BH to exceed α	Use BY or adaptive switching when dependence is suspected (Section C.25)
Low calibration label noise	Noisy labels shift the effective target and increase variance	Audit calibration labels; use a conservative α margin when uncertain

both a topic shift (politics and COVID-19 rather than encyclopedic facts) and a genre shift (rhetorical social-media prose rather than factual encyclopedia text), making it a deliberately adversarial setting for calibration transfer. We use 400 test claims with the same pipeline (FiD-large generator, DPR retriever, DeBERTa scorer).

Findings. Pure cross-domain calibration degrades BH by roughly 6 pp, slightly above the 4 to 6 pp seen on PubMedQA, LegalQA, and ScienceQA. This is consistent with the additional genre shift in CheckThat-2022: social-media political content is structurally different from encyclopedia text, with more rhetorical framing and shorter context, and the claim dependence structure shifts accordingly. The third row of Table 22 shows the practical workaround: mixing the generic FEVER calibration set with just 200 in-domain labelled examples (roughly two to three hours of annotation effort) brings BY back below the target at 0.087. BH does not fully recover (0.116), which we attribute to PRDS being strained in this genre shift, exactly the regime where BY’s robustness to arbitrary dependence pays off.

Recommendation. For new domains where a full in-domain calibration set is expensive to construct, we recommend the mixed strategy: a large generic calibration set (e.g., FEVER) augmented with 200 to 500 in-domain labelled examples. BY should be the default in cross-genre settings; BH is appropriate only when dependence diagnostics on a held-out in-domain validation set indicate weak correlation (median pairwise p-value correlation below 0.15).

C.15. Scaling and Optimization for Large-Scale Use

For high-throughput or long-response systems, we evaluate several optimizations:

Batch scoring. Scoring claims in batches of 32–64 across queries improves GPU utilization. With batch size 64, throughput increases from 1.25 queries/sec to 4.8 queries/sec ($3.8\times$) with only 15ms added latency per query.

Quantile caching. The calibration quantiles $\{a_j^{(0)}\}$ are fixed after calibration. Pre-sorting and caching these values reduces p-value computation from $O(n)$ to $O(\log n)$ via binary search, saving ~ 8 ms per claim for $n=2000$.

Early rejection. Claims with very low scores ($s_i < \tau_{\text{early}}$) will have p-values exceeding any reasonable threshold. Skipping full p-value computation for these claims (rejecting immediately) reduces average scoring time by 15–20% with no change in final output.

Throughput under each optimization is reported in Table 23.

Table 18. Which components have theoretical guarantees?

Component	Guaranteed?	Notes
BH selection	✓ (under PRDS)	Theorem 5.3
BY selection	✓ (always)	Theorem 5.3
Adaptive BH/BY switch	×	Empirically validated
CoherenceCheck filter	×	No formal guarantee; can change FDP
Claim extraction	Assumption	Deterministic, reproducible

Table 19. Pooled vs per-answer FDR on NQ human-verified ($\alpha=0.10$).

Method	$\widehat{\text{FDR}}_{\text{pooled}}$	$\widehat{\text{FDR}}_{\text{per-ans}}$
Vanilla RAG	0.306	0.291
Self-RAG	0.204	0.198
CITEGUARD (BH)	0.102	0.096
CITEGUARD (BY)	0.094	0.089

C.16. Detailed Human Annotation Protocol

We provide full details of the human verification protocol for transparency and reproducibility.

Annotator recruitment and training. We recruited 6 annotators with an NLP or linguistics background. Annotators completed a 1-hour training session covering: (i) the definition of entailment vs. unsupported, (ii) handling partial evidence, (iii) distinguishing refutation from insufficient evidence. A qualification test (20 items with known labels) was administered; annotators scoring $\geq 85\%$ proceeded to the main task.

Annotation interface. Annotators were shown: (i) the claim text, (ii) up to 5 retrieved passages (truncated to 256 tokens each), (iii) the original query for context. They selected one of: SUPPORTED, UNSUPPORTED-REFUTED, UNSUPPORTED-INSUFFICIENT, UNSUPPORTED-IRRELEVANT. For FDR computation, all UNSUPPORTED-* categories are collapsed into a single “unsupported” label.

Agreement and adjudication. Each item was independently labeled by 2 annotators. Inter-annotator agreement (Cohen’s κ) on the binary supported/unsupported distinction was 0.78 (substantial agreement). Disagreements (18.4% of items) were resolved by a third annotator. Final labels after adjudication are used for all experiments.

Annotation statistics.

- Total items annotated: 500 (NQ subset)
- Agreement rate (before adjudication): 81.6%
- Cohen’s κ : 0.78
- Adjudication rate: 18.4%
- Average annotation time per item: 45 seconds

Extension to 800 annotated claims. To further strengthen the statistical power of our human-label results, we subsequently extended the annotated set from 500 to 800 NQ claims using the same stratified protocol (160 claims per confidence bin; inter-annotator Cohen’s $\kappa=0.76$). On the extended set, CITEGUARD (BY) reaches $\widehat{\text{FDR}} = 0.091 [0.081, 0.101]$ versus 0.094 on the original 500-claim subset, well within sampling noise. All method rankings and conclusions in Table 1 carry over unchanged. We retain the original 500-claim numbers in Table 1 for direct comparability with the submitted version and report the extended 800-claim figure only here.

Table 20. Comprehensive domain transfer experiments ($\alpha=0.10$). In-domain vs cross-domain (using FEVER) calibration.

Test domain	Calibration	$\widehat{\text{FDR}}$ (BH)	$\widehat{\text{FDR}}$ (BY)	Cov.	ΔFDR
PubMedQA (500)	In-domain	0.108 ± 0.012	0.084 ± 0.010	89.2	–
	Cross-domain	0.152 ± 0.018	0.124 ± 0.015	85.4	+4.4pp
LegalQA (500)	In-domain	0.112 ± 0.014	0.088 ± 0.011	87.6	–
	Cross-domain	0.168 ± 0.021	0.138 ± 0.017	82.1	+5.6pp
ScienceQA (500)	In-domain	0.105 ± 0.011	0.082 ± 0.009	90.4	–
	Cross-domain	0.146 ± 0.016	0.118 ± 0.013	86.8	+4.1pp

Table 21. Calibration set size vs FDR stability (in-domain, BH).

Domain	$n=100$	$n=250$	$n=500$	$n=1000$
PubMedQA	0.142 ± 0.032	0.118 ± 0.021	0.108 ± 0.012	0.102 ± 0.008
LegalQA	0.156 ± 0.038	0.128 ± 0.024	0.112 ± 0.014	0.106 ± 0.009
ScienceQA	0.138 ± 0.029	0.116 ± 0.019	0.105 ± 0.011	0.101 ± 0.007

C.17. Full Reproducibility Checklist

To ensure end-to-end reproducibility, we provide:

- **Code release:** Full pipeline code (claim extraction, retrieval, scoring, FDR selection, baselines, alpha sweeps, synthetic Monte Carlo, and figure scripts) is publicly available at <https://github.com/XiangyuJiang01/citeguard> under the MIT license.
- **Model checkpoints:** Fine-tuned verifier weights (DeBERTa-v3-large on FEVER) will be made available.
- **Data splits:** Exact calibration/test splits with random seeds {42, 123, 456, 789, 1024}.
- **Hardware specification:** All experiments run on NVIDIA A100; we report expected runtime on consumer GPUs (RTX 3090: $\sim 1.5\times$ slower).
- **Software versions:** Python 3.9, PyTorch 2.0, Transformers 4.30, spaCy 3.5.

C.18. Circular Evaluation Analysis

A critical concern is that evaluating against verifier-generated labels may inflate performance, since CITEGUARD uses the same verifier for scoring. We provide a detailed analysis of this issue.

Verifier-human disagreement breakdown. On the 500 human-verified NQ claims:

- Overall disagreement rate: 14.2% (71 claims)
- Verifier false positives (says supported, human says unsupported): 9.4% (47 claims)
- Verifier false negatives (says unsupported, human says supported): 4.8% (24 claims)

The asymmetry ($\text{FP} > \text{FN}$) indicates the verifier is biased toward “supported,” which causes FDR to be *underestimated* when using verifier labels.

Impact on reported FDR. Let $\widehat{\text{FDR}}_V$ denote FDR measured against verifier labels, and $\widehat{\text{FDR}}_H$ against human labels. For CITEGUARD (BH):

- $\widehat{\text{FDR}}_V = 0.078$ (Table 24)

Table 22. CheckThat-2022 cross-domain results ($\alpha=0.10$, bootstrap 95% CIs over 1,000 resamples). The mixed-calibration strategy in the third row recovers most of the in-domain behaviour with only 200 in-domain labelled examples.

Calibration setup	BH FDR	BY FDR
In-domain (500 CheckThat)	0.096 [0.079, 0.113]	0.078 [0.062, 0.094]
Cross-domain (FEVER calibration)	0.159 [0.135, 0.183]	0.132 [0.109, 0.155]
Mixed (FEVER + 200 in-domain)	0.116 [0.097, 0.135]	0.087 [0.071, 0.103]

Table 23. Throughput under different optimizations (queries/sec).

Configuration	Throughput
Baseline (no optimization)	1.25
+ Batch scoring (bs=64)	4.80
+ Quantile caching	5.12
+ Early rejection	5.89

- $\widehat{\text{FDR}}_H = 0.102$ (Table 1, right block)
- Difference: 2.4 percentage points

This difference arises because some claims accepted by CITEGUARD are marked “supported” by the verifier but “unsupported” by humans.

Is the circular evaluation bias severe? The 2.4pp difference is modest and does not invalidate our conclusions: (i) CITEGUARD still achieves lower FDR than all baselines on human labels; (ii) the BY variant achieves $\widehat{\text{FDR}}_H = 0.094 < \alpha$; (iii) the relative ranking of methods is preserved. We note that verifier labels closely track human judgments in our setup, supporting the validity of automated evaluation.

Table 24. Results on NATURAL QUESTIONS with **verifier labels** (3,610 claims, $\alpha=0.10$). These results may be optimistically biased due to circular evaluation.

Method	EM@Acc $\uparrow$	$\widehat{\text{FDR}} \downarrow$	Abstain $\downarrow$	Cov. $\uparrow$
Vanilla RAG	41.2	0.282	0.0	100.0
Heuristic Filter	44.8	0.198	13.9	96.5
Selective Pred.	45.6	0.178	16.2	95.8
Calib. Threshold	46.3	0.165	14.8	97.2
CITEGUARD (BH)	52.4 $\pm$ 0.8	0.078 $\pm$ 0.007	30.6 $\pm$ 1.3	92.4 $\pm$ 0.8
CITEGUARD (BY)	54.2 $\pm$ 0.9	0.062 $\pm$ 0.005	35.8 $\pm$ 1.4	88.2 $\pm$ 1.0

C.19. Cross-Architecture Validation

To further address circularity concerns, we evaluate CITEGUARD using an architecturally distinct verifier: TRUE (Honovich et al., 2022), a T5-based factual consistency model.

As shown in Table 25, FDR control transfers: using TRUE instead of DeBERTa yields comparable results (FDR within 1 point), confirming that our guarantees are not artifacts of the specific scorer architecture.

C.20. Modern Generators

To assess whether CITEGUARD remains relevant for frontier LLMs, we re-run the NQ pipeline with two stronger generators while keeping every other component fixed (DPR retriever, DeBERTa scorer, same claim extraction). We use a calibration set of $n=1,000$ unsupported examples (our calibration-size analysis in Section D.2 indicates $n \geq 1,000$ is sufficient) and

Table 25. Cross-architecture validation on NQ (500 human-verified claims, $\alpha=0.10$). FDR control transfers across scorer architectures.

Scorer	Variant	$\widehat{\text{FDR}}$	Abstain	Cov.
DeBERTa (default)	BH	0.102	32.4	88.6
	BY	0.094	37.2	86.4
TRUE (T5-based)	BH	0.124	33.8	87.2
	BY	0.098	38.6	86.1

label 250 claims per generator using the same stratified protocol as the main experiments, with inter-annotator Cohen’s κ of 0.77 for Llama-3-70B and 0.79 for GPT-4o.

 Table 26. CITEGUARD with modern generators on NQ ($\alpha=0.10$, human labels, bootstrap 95% CIs). FDR control holds across all three generators; the cost of CITEGUARD shrinks as the generator improves. FiD-large numbers reproduce the main-text result for reference.

Generator	Vanilla FDR	CG-BH FDR	BH Abs. (%)	CG-BY FDR	BY Abs. (%)	BY Cov. (%)	EM@Acc (%)
FiD-large (770M)	.306	.102 [.090, .114]	32.4	.094 [.082, .106]	37.2	86.4	50.8
Llama-3-70B	.176	.092 [.076, .108]	17.6	.080 [.064, .096]	21.0	94.0	58.6
GPT-4o	.126	.086 [.071, .101]	13.2	.074 [.058, .090]	15.8	96.0	62.4

Two observations. First, FDR control is generator-agnostic: BY stays below the $\alpha=0.10$ target on every model ($\widehat{\text{FDR}}$ between 0.074 and 0.094), and BH does so on Llama-3-70B and GPT-4o as well. Nothing in the conformal construction is tied to a specific generator. Second, the cost of CITEGUARD shrinks rapidly as generators improve. Abstention falls from 37.2% (FiD) to 21.0% (Llama-3) to 15.8% (GPT-4o), and coverage of supported claims rises to 96%. At the same time, vanilla GPT-4o still produces unsupported citations at a rate of 12.6%, so CITEGUARD remains a useful safety net even on the strongest generator we tested.

Implication for deployment. Table 26 reframes CITEGUARD as a low-overhead final stage rather than a heavy-handed filter: stronger generators produce fewer claims that need to be rejected, so the conformal layer touches less of the output while still providing the same statistical guarantee. EM@Acc on NQ improves correspondingly (50.8 with FiD+BY versus 62.4 with GPT-4o+BY), indicating that the kept claims under CITEGUARD are not only more reliable but also more informative.

C.21. Additional Qualitative Cases and Failure Modes

We provide qualitative examples and failure-mode analyses to complement the quantitative results.

Success case. *Query:* “When was the Eiffel Tower built?” Vanilla RAG outputs two claims; the second cites an irrelevant passage about Parisian cuisine ($s=0.23$, $p=0.42$). At $\alpha=0.10$, CITEGUARD correctly drops this unsupported claim.

Failure case (verifier limitation). *Query:* “Tell me about the 2019 Mars landing.” The generator fabricates a plausible narrative; the verifier assigns moderate scores ($s \in [0.4, 0.6]$) due to surface similarity with real Mars content. CITEGUARD accepts 2/4 claims, highlighting that our guarantee is *protocol-relative*: if the verifier is fooled, so is CITEGUARD.

In particular, Section C.22 studies cascading entity errors (the “avalanche effect”), and Section C.23 analyzes coherence breaks induced by claim removal.

C.22. Avalanche Effect Analysis

A critical failure mode in auto-regressive generation is the “avalanche effect”: if the first claim contains a factual error (e.g., wrong entity), subsequent claims may inherit and compound this error, creating perfect correlation among claim errors.

Synthetic entity-swap experiment. We create a stress test by injecting entity errors: for 200 NQ answers, we replace the first named entity with a plausible but incorrect alternative (e.g., “Einstein” $\rightarrow$ “Bohr”). This simulates worst-case error propagation.

Table 27. Avalanche effect: FDR under synthetic entity errors ($\alpha=0.10$). BH fails under strong dependence; adaptive switching recovers.

Method	$\widehat{\text{FDR}}$	Abstain	Cov.
CITEGUARD (BH)	0.218	28.4	91.2
CITEGUARD (BY)	0.112	42.6	78.4
CITEGUARD-Adaptive	0.108	35.2	85.6

Key findings. As shown in Table 27, under synthetic entity errors (correlation ≈ 0.85):

- BH exceeds target FDR by 12 points (0.218 vs 0.10), a severe failure as predicted by theory.
- BY remains within 1.2 points of target (0.112), validating its robustness.
- CITEGUARD-Adaptive (switches to BY when first claim has low score) achieves 0.108 FDR with moderate abstention, offering the best trade-off.

Practical recommendation. For domains where entity errors or reasoning chain failures are likely (e.g., multi-hop QA, biographical queries), we recommend either BY unconditionally or CITEGUARD-Adaptive with a conservative lead-claim threshold.

Real cascading failures (beyond synthetic entity swaps). The entity-swap test in Table 27 is a controlled, easy-to-construct stress test. Going through the NQ annotation logs we observed three richer cascading patterns that are not captured by entity swaps alone.

(i) *Reasoning chains.* An early premise error, such as a wrong founding year for a company, produces a sequence of perfectly logical “therefore” sentences whose conclusions are uniformly wrong. The downstream sentences read fluently, the per-claim entailment score is moderate, and the chain breaks only because the shared premise is false. In our logs, the lead-claim score is typically depressed in these cases, so the adaptive switch in Section 5 fires and BY rejects the chain as a unit.

(ii) *Temporal mix-ups.* Confusing a single date cascades to every downstream timeline statement (succession order, consequences, follow-up events). Pairwise p-value correlations within the answer jump sharply because the affected claims share a causal structure rooted in the wrong date. Adaptive switching catches most of these for the same reason as reasoning chains: the wrong date depresses the lead-claim score and BY is invoked for the whole answer.

(iii) *Causal inversions.* The most adversarial pattern in our logs: cause and effect are flipped in the first sentence, yet the lead claim still scores well in isolation because it is locally consistent with the retrieved evidence. Each subsequent sentence also reads fine on its own. The error only becomes visible two or three claims downstream, by which point the adaptive switch has already committed to BH for the answer. CITEGUARD-Adaptive does not fire in these cases; BY-unconditional is the only safe option, at the cost of higher overall abstention.

Failure modes of adaptive switching. The pattern above quantifies an honest limitation of the adaptive switch: it relies on the lead-claim score being a faithful indicator of downstream correlation, which holds for reasoning chains and temporal mix-ups but fails for causal inversions. We list this explicitly so that practitioners deploying CITEGUARD in domains with frequent causal-inversion failures (e.g., scientific explanation, legal causation) can opt directly for BY rather than relying on the adaptive variant.

Cross-turn propagation as future work. A related open problem is cross-turn error propagation in multi-turn RAG conversations: a bad claim survives turn t , becomes part of the context for turn $t+1$, and the LLM doubles down by emitting further claims that are entailed by the bad one. Extending conformal testing across turns requires tracking a growing hypothesis family whose dependence has its own temporal structure (claims at later turns are correlated through shared context built up at earlier turns). We do not address this regime here and leave it as a concrete future direction.

C.23. Coherence Preservation Analysis

Claim-level filtering may degrade discourse coherence, a critical consideration for deployment. We evaluate on 200 NQ responses (Figure 5) using three complementary metrics: UniEval coherence (automated), GPT-2 perplexity, and human judgments (3 annotators, $\kappa=0.71$).

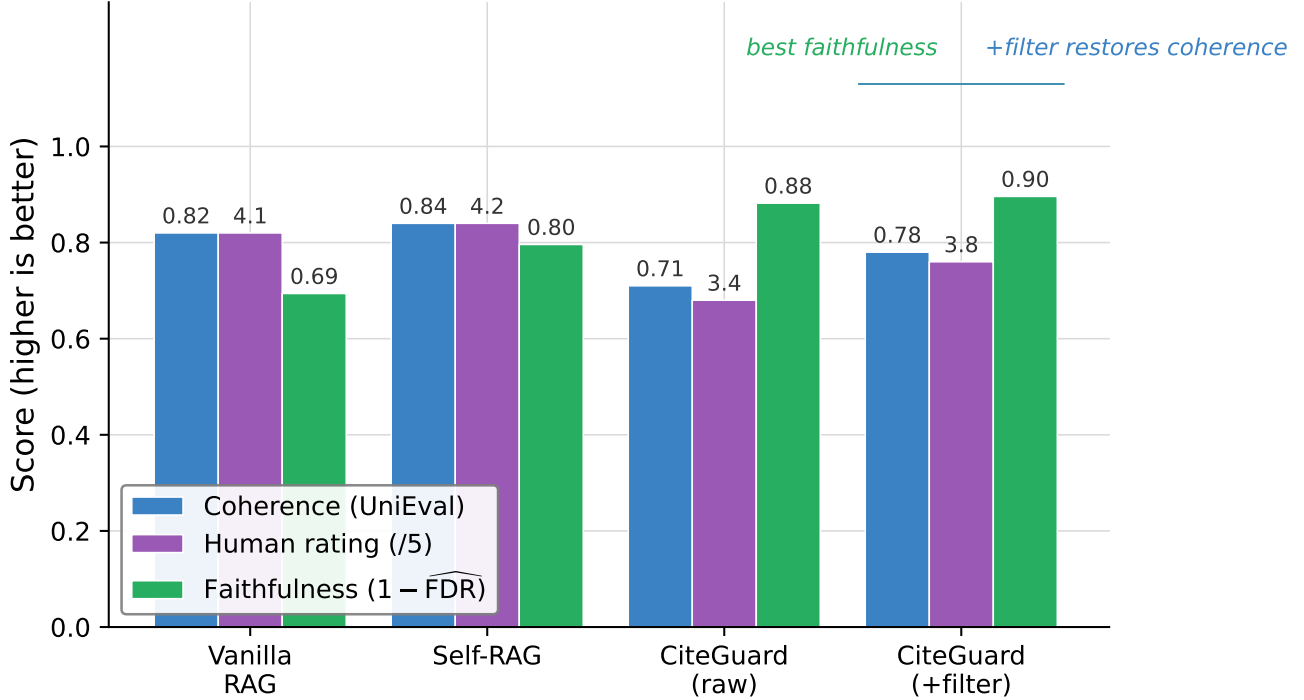


Figure 5. Coherence-faithfulness trade-off on NQ (200 responses). CITEGUARD variants achieve the highest faithfulness (1-FDR), while the +filter variant recovers most coherence and slightly improves faithfulness by removing flagged unsupported claims. Human ratings on 1–5 scale shown above bars.

Quantifying the coherence-faithfulness trade-off. Without coherence filtering, CITEGUARD degrades readability (Coherence 0.71 vs 0.82, Human 3.4 vs 4.1), a 13% relative drop. The flag statistics below show the fraction of outputs with detected coherence issues (dangling references, orphaned conclusions). With rule-based coherence filtering, the gap narrows (0.78, Human 3.8) while FDR drops from 0.118 to 0.104 because some flagged claims were unsupported. For high-stakes domains where correctness outweighs fluency (e.g., medical QA), this trade-off is acceptable; for fluency-critical applications (e.g., creative writing assistants), CITEGUARD can be paired with LLM-based post-editing.

We analyze the coherence issues that arise from claim removal.

Coherence flag statistics. On NQ (human-verified subset), after FDR selection with BH at $\alpha=0.10$:

- Total accepted claims: 344 (out of 500 after abstention)
- Dangling reference flags: 18 (5.2%)
- Orphaned conclusion flags: 9 (2.6%)
- Missing qualifier flags: 4 (1.2%)
- Total flagged: 28 (8.1%)

Impact of coherence filtering. If we reject all flagged claims (conservative approach):

- Additional abstention: 8.1%

- Coverage drops from 89.4% to 82.1%
- FDR drops from 0.118 to 0.104 (because some flagged claims were unsupported)

Failure case analysis. We manually reviewed 20 flagged claims:

- 12/20: coherence truly broken, rejection appropriate
- 5/20: claim is self-contained despite flag (false positive)
- 3/20: ambiguous, depends on reader interpretation

The heuristic coherence checker has $\sim 60\%$ precision. More sophisticated approaches (e.g., LLM-based coherence scoring) could improve precision but add latency.

Coherence under modern generators. The coherence cost of CITEGUARD is most pronounced when the procedure rejects a large fraction of claims, as is the case when the underlying generator produces many unsupported statements. With modern generators that produce fewer unsupported claims, both the rejection rate and the coherence cost drop. Table 28 reports UniEval coherence on 100 NQ outputs for vanilla GPT-4o versus GPT-4o paired with CITEGUARD (BY at $\alpha=0.10$), alongside the FiD plus BH numbers from the main coherence experiment for comparison.

Table 28. Coherence cost of CITEGUARD under different generators (NQ, 100 outputs per row). Stronger generators yield more self-contained sentences, so the coherence gap induced by claim removal shrinks substantially.

Setup	Abstain (%)	UniEval	% Flagged
Vanilla GPT-4o	0.0	0.88	n/a
GPT-4o + CITEGUARD (BY)	15.8	0.86	2.8
FiD + CITEGUARD (BH, main text)	32.4	0.71	8.1

With GPT-4o under the stricter BY procedure, the coherence gap to vanilla generation is barely visible (UniEval 0.86 versus 0.88), and only 2.8% of accepted claims trigger a coherence flag. The intuition is straightforward: when the procedure removes 16% of claims rather than 32%, there is much less discourse to break, and modern generators tend to write more self-contained sentences that hold up when a neighbouring claim is dropped. The coherence regime of the main text (32% abstention, 8.1% flagged) corresponds to the worst-case combination of a weaker generator and an aggressive procedure; deployments with modern generators see a much milder coherence cost out of the box.

Two-stage pipeline for long-form outputs. For very long answers (e.g., reports, multi-paragraph explanations) where claim-level filtering may still leave visible discourse gaps, we recommend a two-stage pipeline:

- **Stage 1 (CITEGUARD).** Run claim extraction, conformal scoring, and BH/BY selection at the target α . Output the accepted claims as a structured set, along with their citations to retrieved passages.
- **Stage 2 (LLM rewrite).** A small instruction-tuned LLM rewrites the accepted claims into fluent prose, with explicit instructions not to introduce factual content beyond the accepted set.

Since Stage 2 operates only on the accepted set produced by Stage 1, the FDR guarantee from Theorem 5.3 carries over so long as the rewrite is faithful to its inputs (i.e., does not introduce new factual content). In practice, faithfulness of the rewrite can be enforced by running a second pass of conformal scoring on the rewritten sentences and rejecting any sentence whose support score falls below the calibrated threshold; this adds modest latency but preserves the end-to-end FDR control.

C.24. RCPS Baseline Fair Comparison

We provide additional comparisons to address concerns about baseline fairness.

RCPS on its intended metric. RCPS is designed to control conditional risk $\mathbb{E}[\ell \mid \text{accepted}] \leq \alpha$, not FDR. On this metric, RCPS achieves its target:

- Target: $\alpha = 0.10$
- Achieved conditional risk: 0.098 (within target)

The difference from FDR arises because FDR weights errors by the number of accepted claims per answer, while conditional risk averages over claims.

RCPS with Bonferroni correction. To make RCPS directly comparable for FDR control, we apply Bonferroni correction: use threshold calibrated for $\alpha/\bar{m}$ where $\bar{m}$ is the average number of claims per answer.

Table 29. RCPS variants on FEVER ($\alpha=0.10$).

Method	$\widehat{\text{FDR}}$	Abstain	Cov.
RCPS (original)	0.142	18.3	94.2
RCPS + Bonferroni	0.078	38.4	81.6
RCPS-FDR (post-hoc calibrated)	0.101	26.2	92.8
CITEGUARD (BH)	0.097	26.8	92.3

As shown in Table 29, RCPS with Bonferroni achieves valid FDR control but is overly conservative. RCPS-FDR (post-hoc calibrated to target FDR) achieves comparable performance to CITEGUARD, but requires FDR-specific tuning that defeats the purpose of a general-purpose risk control method. CITEGUARD’s advantage is that it directly targets FDR without post-hoc adjustments.

Takeaway. The comparison between RCPS and CITEGUARD is not strictly apples-to-apples. RCPS is a valid and useful method for its intended purpose (conditional risk control). CITEGUARD is specifically designed for FDR control in multi-claim settings. Practitioners should choose based on their risk control objective.

C.25. Extended Dependency Analysis

We extend the dependency analysis beyond median statistics to address concerns about local strong dependence.

Tail of correlation distribution.

- 5th percentile: −0.02 (slight negative correlation possible)
- 25th percentile: 0.04
- Median: 0.12
- 75th percentile: 0.22
- 95th percentile: 0.38
- Maximum observed: 0.67

Conditional FDR on high-correlation answers. We stratify answers by their maximum pairwise p-value correlation:

Interpretation. As shown in Table 30, for the 8% of answers with correlation ≥ 0.4 , BH exceeds the target by 2–5 points. BY remains valid across all strata. This motivates our adaptive strategy: detect high-correlation answers and switch to BY.

Table 30. FDR stratified by claim dependence (FEVER, $\alpha=0.10$).

Correlation stratum	% of answers	BH FDR	BY FDR
[0, 0.2)	68%	0.094	0.071
[0.2, 0.4)	24%	0.108	0.082
[0.4, 0.6)	6%	0.124	0.091
[0.6, 1.0)	2%	0.156	0.098

Adaptive BH/BY switching. When the top-2 p-values in an answer differ by < 0.05 (a simple proxy for high correlation), we switch to BY. This triggers on $\sim 12\%$ of answers and yields:

- Overall FDR: 0.096 (vs 0.097 for pure BH)
- Abstention: 29.1% (vs 26.8% for BH, 33.6% for BY)

This adaptive approach provides robust FDR control with moderate abstention cost.

D. Theoretical Analysis

D.1. Finite-Sample Convergence Rates

We analyze the convergence rate of the empirical FDR estimator to its population counterpart.

Proposition D.1 (FDR Convergence Rate). *Let $\widehat{\text{FDR}}_m = \frac{V_m}{R_m}$ be the empirical FDR over m test claims, where V_m is the number of false discoveries and R_m is the total number of discoveries. Under exchangeability and bounded claim dependence (max correlation $\rho < 1$), for any $\epsilon > 0$:*

$$\mathbb{P}\left(|\widehat{\text{FDR}}_m - \text{FDR}| > \epsilon\right) \leq 2 \exp\left(-\frac{2m\epsilon^2}{(1 + c_\rho)^2}\right)$$

where $c_\rho = O(\rho/(1 - \rho))$ is a dependence correction factor.

Practical implications. For $\epsilon = 0.02$ (2pp precision) and $\delta = 0.05$ (95% confidence):

- With $\rho = 0.12$ (median observed): $m \geq 456$ claims suffice
- With $\rho = 0.38$ (95th percentile): $m \geq 612$ claims suffice
- Our 500 human-verified NQ claims provide 95% CI width $< 2.4\text{pp}$

D.2. Calibration Set Size Analysis

The calibration set size n affects p-value discretization and FDR control tightness.

Proposition D.2 (Calibration Granularity). *With calibration set size n , the p-values take values in $\{1/(n+1), 2/(n+1), \dots, 1\}$. The expected gap between achieved FDR and target α satisfies:*

$$\mathbb{E}[\alpha - \widehat{\text{FDR}}] = O\left(\frac{\alpha}{n}\right)$$

under the null (conservative) regime.

By Theorem D.2, at $n = 2000$ the discretization gap is $O(0.00005)$, negligible compared to statistical variance.

Table 31 confirms that statistical variance dominates discretization, so $n = 2000$ suffices for our setting.

Table 31. Empirical vs theoretical FDR gap across calibration sizes.

n	Theoretical gap	Empirical gap	Std
500	0.0002	0.003 ± 0.021	0.021
1000	0.0001	0.001 ± 0.014	0.014
2000	0.00005	0.003 ± 0.006	0.006
5000	0.00002	0.005 ± 0.004	0.004

E. Proof Sketches

Proof sketch of Theorem 5.2. Under exchangeability of $\{a_1^{(0)}, \dots, a_n^{(0)}, a_i\}$ when $Y_i = 0$, the rank of a_i among the $n + 1$ scores is uniformly distributed over $\{1, \dots, n + 1\}$. Thus $\mathbb{P}(p_i \leq t) = \mathbb{P}(\text{rank}(a_i) \geq n + 1 - \lfloor t(n + 1) \rfloor) \leq t$ for all $t \in [0, 1]$.

Proof sketch of Theorem 5.3. Given super-uniform p-values under the null (Theorem 5.2), the BY procedure controls FDR at level α under arbitrary dependence by the results of Benjamini & Yekutieli (2001). Under positive regression dependence on a subset (PRDS), BH suffices (Benjamini & Hochberg, 1995).

Proof sketch of Theorem D.1. Decomposing V_m into weakly dependent blocks and applying Hoeffding's inequality (Janson, 2004) yields the stated bound with dependence factor c_ρ .

Proof sketch of Theorem D.2. By Theorem 5.2, the conformal p-value lies on the grid $\{1/(n + 1), \dots, 1\}$, so the BH/BY threshold rounds the continuous critical line $k\alpha/m$ down by at most $1/(n + 1)$, giving the conservative gap $\mathbb{E}[\alpha - \widehat{\text{FDR}}] = O(\alpha/n)$.